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System Requirements:

On-Course Evaluation Analysis System requires the use of Microsoft Access 2000 database software and Microsoft Excel for advanced reports, therefore the system requirements are only those needed to install and run these programs.

It is recommended that the user have the following system requirements:

- Windows 95 or Higher
- 16 Megabytes of RAM or Higher and a Pentium Processor
- At least 12 Megabytes of Free Space on Hard Drive

Installation Instructions:

Floppy Disk Installation:

1. Create Directory C:\OnCourse
2. Go to the Windows Start Button
3. Select Run from the Start Menu
4. Type in a:
5. Open the self-extracting .zip file OnCourse.exe to C:\OnCourse
6. Copy the file MSVBM60.dll into the C:\Windows\System directory
7. Double Click on OnCourse.exe to launch program

A CD-ROM installation disk is available upon request



Project of:

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At the Request of:

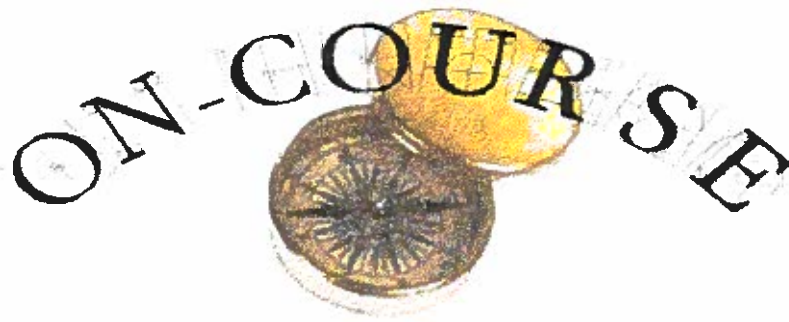
Dr. Bill Bowers
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Charleston Southern University
(843) 863-7908

Under the Supervision of:

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(843) 863-8083

Special Thanks to:

Mr. Robert Thurston
Mr. Todd Lea



Purpose:

On-Course uses Q-Sort Methodology (McKeown & Thomas, 1988) to evaluate University courses by automating the analysis of the evaluations, store and manipulate the evaluation data, and generate valuable reports on progress.

Background:

Charleston Southern University, along with many other institutions, has a system to evaluate the courses that are offered as well as the instructors that are teaching them. These evaluations serve to monitor the progress of the subject being evaluated to ensure optimum results for the students attending the university.

Currently at Charleston Southern University, the system of evaluations provides minimal information about the course, and does not coincide with a standard that is desired for each course and instructor. On-Course implements a system that can get data from student evaluations, fit it into a bell curve for easy analysis, store the data in an extensive database, and then generate reports based on how each course and instructor is performing as compared to the ideal standard.

**REQUEST FOR
INFORMATION
SYSTEM SERVICES**

DATE OF REQUEST	SERVICE REQUESTED FOR DEPARTMENT(S)
October, 14 2000	Business Department
SUBMITTED BY (key user contact)	EXECUTIVE SPONSOR (funding authority)
Name Dr. Bill Bowers	Name
Title Business Professor	Title
Office Jones Hall	Office
Phone (843) 863-7908	Phone

TYPE OF SERVICE REQUESTED:

- | | |
|--|---|
| ☐ Information Strategy Planning | ☐ Existing Application Enhancement |
| ☒ Business Process Analysis and Redesign | ☐ Existing Application Maintenance (problem fix) |
| ☒ New Application Development | ☐ Not Sure |
| ☐ Other (please specify _____) | |

BRIEF STATEMENT OF PROBLEM, OPPORTUNITY, OR DIRECTIVE (attach additional documentation as necessary)

Charleston Southern University currently has a system of course evaluations that allows students to evaluate the course that they are taking as well as the professor who is instructing the course. The information gained through these evaluations is very valuable to the professor, the department, and to the overall academic success of the university. This information can be used to see which courses and professors are succeeding in their work and which could use improvement. The problem with the current system is that there are no ideal standards that can be used to compare each class to the ideal classroom situation as seen by the university and the members of each department. Dr. William Bowers has requested a system be created that will allow students to evaluate each course and professor and then compare the results to an ideal scenario that has been previously established. With the new system, the data will be collected in such a way that it provides various statistical analysis reports that can give the university and the departments much more information than they have yet been able to receive from their current system of course evaluations.

BRIEF STATEMENT OF EXPECTED SOLUTION

The new system of course evaluations will use the Q Methodology (McKeown & Thomas, 1988) to collect and analyze students' responses. A set number of statements will be given and the student distributes the statements throughout several numbered columns whether or not the statement is most like or least like the class or professor that they are evaluating. The data will form a bell curve that can then be inputted into the computer system that will store the data in a database and create useful reports of information for the university and the department to evaluate. From these reports it will be clear to see how each class or professor compares to the ideal classroom setting that has been established by Charleston Southern University.

ACTION (ISS Office Use Only)☒ Feasibility assessment approvedAssigned to Karen Thurston☐ Feasibility assessment waived

Approved Budget \$ _____

Start Date 10/14/00 Deadline 5/9/01☐ Request delayed

Backlogged until date: _____

☐ Request rejected

Reason: _____

Authorized Signatures:

Chair, ISS Executive Steering Body_____
Project Executive Sponsor



PROBLEM STATEMENTS

PROJECT: Student Course Evaluation Analysis	PROJECT MANAGER: Karen Thurston
CREATED BY: Karen Thurston	LAST UPDATED BY: Karen Thurston
DATE CREATED: 10/14/00	DATE LAST UPDATED: 10/14/00

Brief Statements of Problem, Opportunity, or Directive	Urgency	Visibility	Annual Benefits	Priority or Rank	Proposed Solution
1. Current system does not compare courses or professors to a set standard.	ASAP	High	Will set a standard for all classes.	1	Use Q Methodology that will have an ideal evaluation that is desired of the course and compare students' responses to the ideal.
2. Data needs to be processed using statistical analysis to provide useful information to the university and the departments.	4 months	High	Will provide detailed analysis of the data.	2	Use the data, which forms a bell curve, and analyze it using several statistical equations.
3. Need an automated system to analyze course evaluations.	7 months	High	Will speed process and give accurate results with less effort.	3	Create automated system that will process the data and generate reports.

FIGURE 4.8



Project Feasibility Assessment Report

I. Executive Summary

A. Summary of Recommendation

On September 15, 2000, Dr. Bill Bowers contacted Greenstone Consulting in regards to a project that would greatly enhance the current system of student course evaluations at Charleston Southern University. Greenstone Consulting will, using a new method of evaluation, create an automated system that will store evaluation data into a database and create reports of the data gathered from the evaluations. The new system will use Q Methodology to compare the students' views of the course and the professor to that of the university's ideal of what the course and professor should be. The new system will take this data, allow it to be inputted for each student, automatically store it in a database, and produce reports that will contain valuable information to the university and each department.

B. Brief Statement of Anticipated Benefits

This new system, once implemented, will provide immediate benefits not only to the university, but also to each individual department, each professor, and possibly even to the students. The data will be analyzed to compare how different the course is to the ideal set by the university. It will also show how different courses compare, and which methods and styles of teaching prove most effective to meeting the university's standard. The major benefit will be that problems and successes can be easily spotted and trouble areas can be fixed for better overall academic success.

C. Brief Explanation of Report Contents

The contents of this report following this summary include a detailed description of the Student Course Evaluation project. It will include Greenstone Consulting, along with Dr. Bill Bowers, recommendations and ideas to solve the problems involved in the project. This report will also contain the project plan and schedule of each step of the project.

II. Background Information

A. Brief Description of Project Request

Dr. Bill Bowers of Charleston Southern University has requested of Greenstone Consulting that an automated computer system be created that can take the students evaluations and produce useful reports for the university and the individual departments. The current system is not set up to compare each course and professor to a standard set by the university, and therefore does not give as accurate of information about where the weaknesses are and which areas need most improvement. The statistics involved with the new system would give the most accurate information possible that could be used for improvements of the courses as well as the professors. The goal is to have the system complete by May of 2001. The goal of this system is to make the courses and operations at Charleston Southern University more effective by providing an accurate evaluation of each course and professor by the students of the university.

B. Brief explanation of the survey phase activities

Greenstone Consulting met with Dr. Bill Bowers of Charleston Southern University in September 2000. Dr. Bowers will be the main contact for the project, and Karen Thurston will be the project manager for Greenstone Consulting. Karen will work with the assistance of Mr. Jim Roberts to design and implement this system as requested by Dr. Bill Bowers. The request was made by Dr. Bowers and evaluated by Karen Thurston. After evaluating the problems, opportunities, and objectives of this request, it has been decided by Greenstone Consulting to go forth with this request. An allotted time of eight months will be given to the completion of this project, with the final deadline being May, 2001.

III. Detailed Recommendation

A. Narrative Recommendation

1. Immediate fixes

Due to the fact that Student Course Evaluations are only performed once a semester, the university will continue with their current system of evaluating courses until the new system is completed and adopted by the university. Until that time there will be no immediate fixes to the system.

2. Enhancements

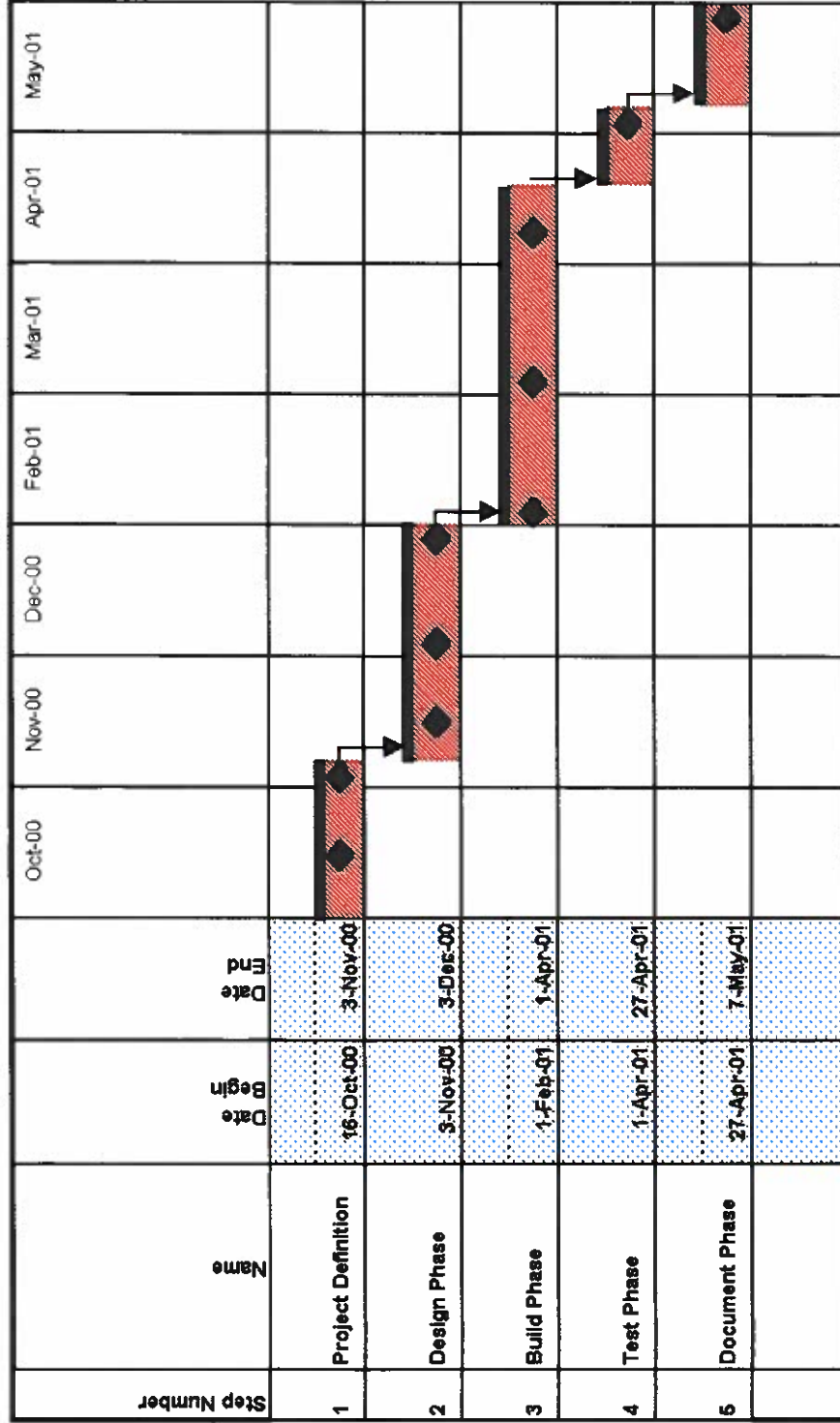
The enhancements made to the old system will be that of using Q Methodology to evaluate the courses instead of only a few questions answered using 1-5 to rate most like or least like the course. The Q Methodology will use 32 statements that will be divided amongst 9 categories, forcing the answers into a bell curve.

3. New Systems Development

Currently there is no automated way of reporting the results of the student evaluations. The new system will provide several options of analyzing the data, including sorting data by the mean, calculating the correlation between different classes, and various other ways. The new system will produce reports from the input that will reflect accurately the overall perception of the course and professor by the students that are enrolled in that course.

B. Project Plan

- 1. Initial Master Project Plan (Phase Level - Gantt Chart)**
- 2. Detailed Plan for the Study or Definition Phase (Gantt Chart)**



Step Number	Project Definition and Design Phase	Begin Date	End Date	Oct-00	Nov-00	Dec-00
	Request for Systems Services					
1	Problem Statement Matrix	16-Oct-00	23-Oct-00			
2	Feasibility Assessment Report	16-Oct-00	23-Oct-00			
3	Gantt Charts Prob, Opp, Obj & Constr Matrix	23-Oct-00	5-Nov-00			
4	E-R Diagram Entity/Definition Matrix	5-Nov-00	27-Nov-00			
5	Data Flow Diagrams E-R List & Decision Tables	28-Nov-00	1-Dec-00			
6	Locatin Connectivity Diagram Candidate & Feasibility Matrix	28-Nov-00	1-Dec-00			
7	Systems Proposal Report	1-Dec-00	10-Dec-00			



PROBLEMS, OPPORTUNITIES, OBJECTIVES AND CONSTRAINTS MATRIX

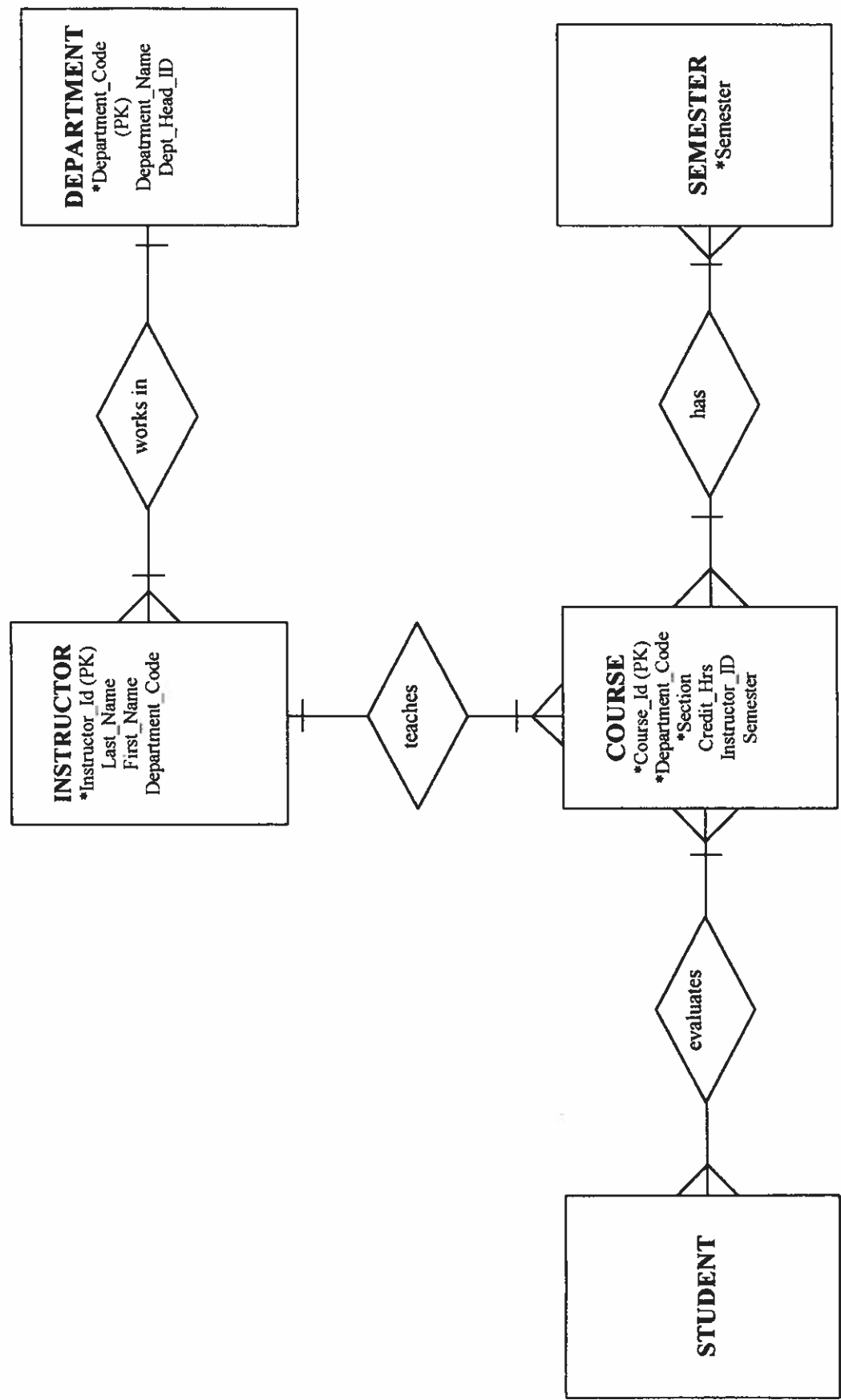
Project:	Student Course Evaluation Analysis	Project Manager:	Karen Thurston
Created by:	Karen Thurston	Last Updated by:	Karen Thurston
Date Created:	10/22/00	Date Last Updated:	10/22/00

CAUSE AND EFFECT ANALYSIS		SYSTEM IMPROVEMENT OBJECTIVES	
Problem or Opportunity	Causes and Effects	System Objective	System Constraint
1. There is no automated program to generate reports from data collected by student course evaluations. 2. The current system of evaluations is not very thorough and does not give enough information to derive detailed reports.	1. The current system of evaluations uses a basic form with only a few questions. 2. The data received by these evaluations is not in a format that can be used for simple statistical calculations. 3. Reports need to be generated that give the university and the departments solid data on each course and each professor, as seen by the students.	1. To use a new system of course evaluations that will implement Q Methodology. 2. To retrieve the data from these evaluations and perform statistical calculations on this data. 3. To create a system to automate the process of analyzing the results. 4. To produce reports that will reflect the students' views on each course and professor compared to the benchmark established by the university.	1. The automated system must eventually be able to be multi-user 2. The system should be user friendly and provide an input screen and several report choices based on the input. 3. Desired completion date of May 2001.

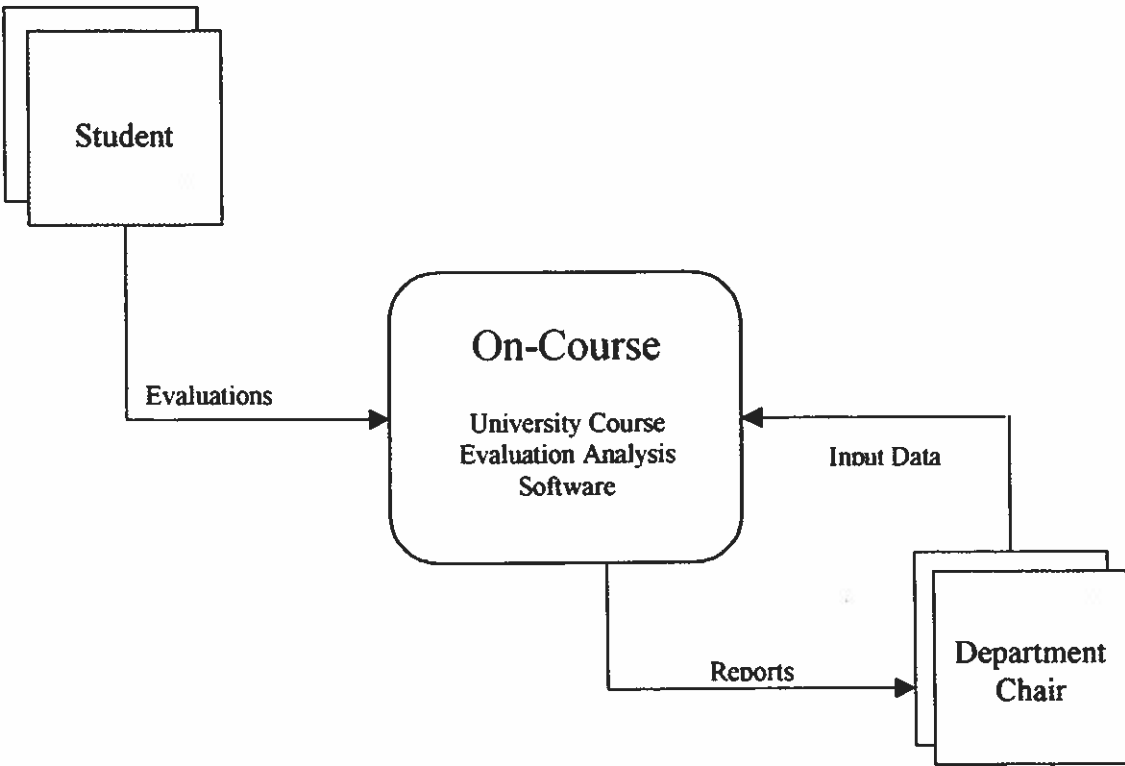
Entity/Definition Matrix

ENTITY	BUSINESS DEFINITION
STUDENT	A person enrolled in a course at the university.
COURSE	A course taught by the professor that has been approved by the university.
PROFESSOR	An employee of the university who works in a department and teaches a course.
SEMESTER	A period of time during which a course is offered by the university.
DEPARTMENT	A department is established by the university and is comprised of professors who teach courses of a particular subject matter.

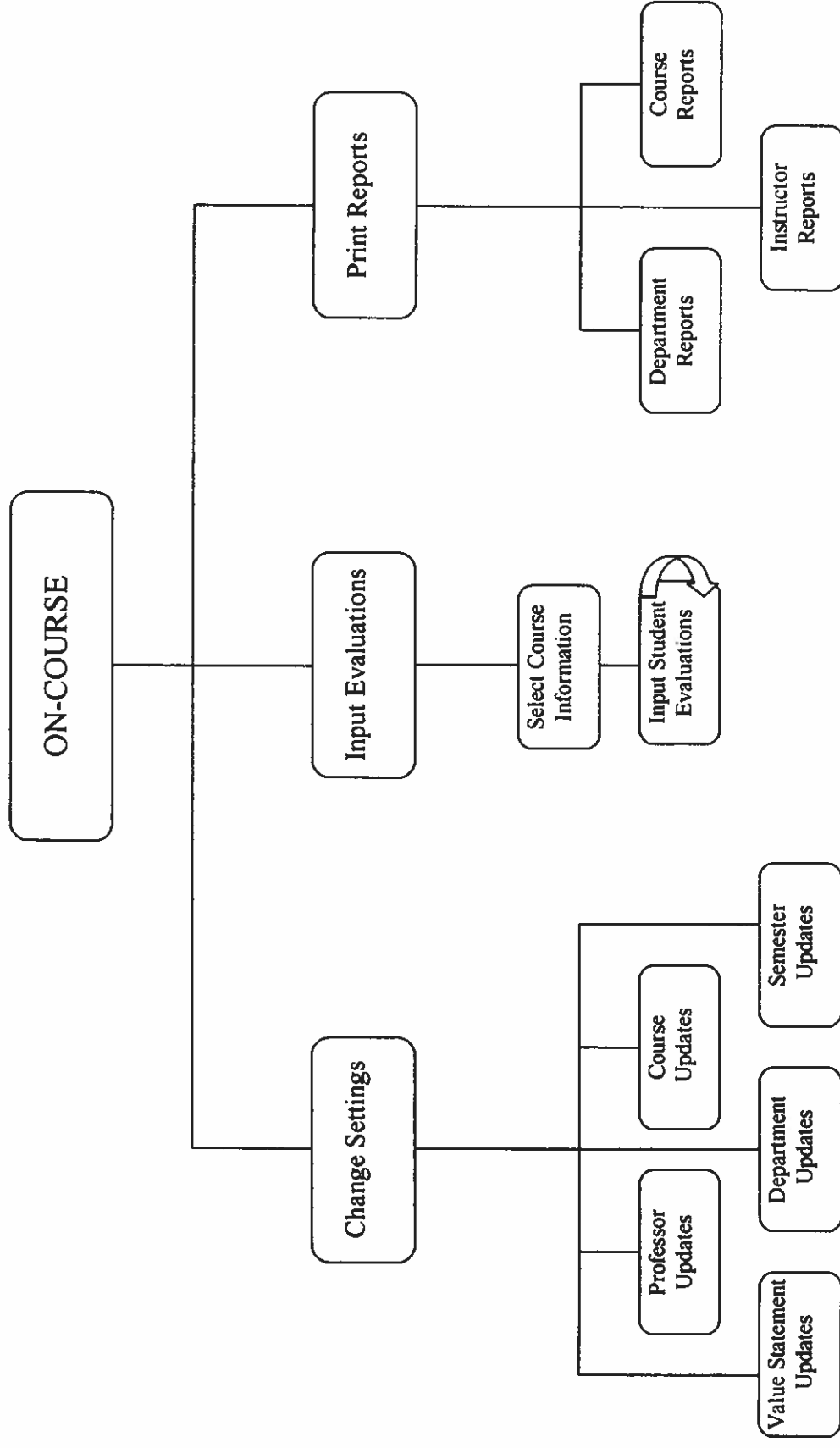
Entity Relationship Diagram



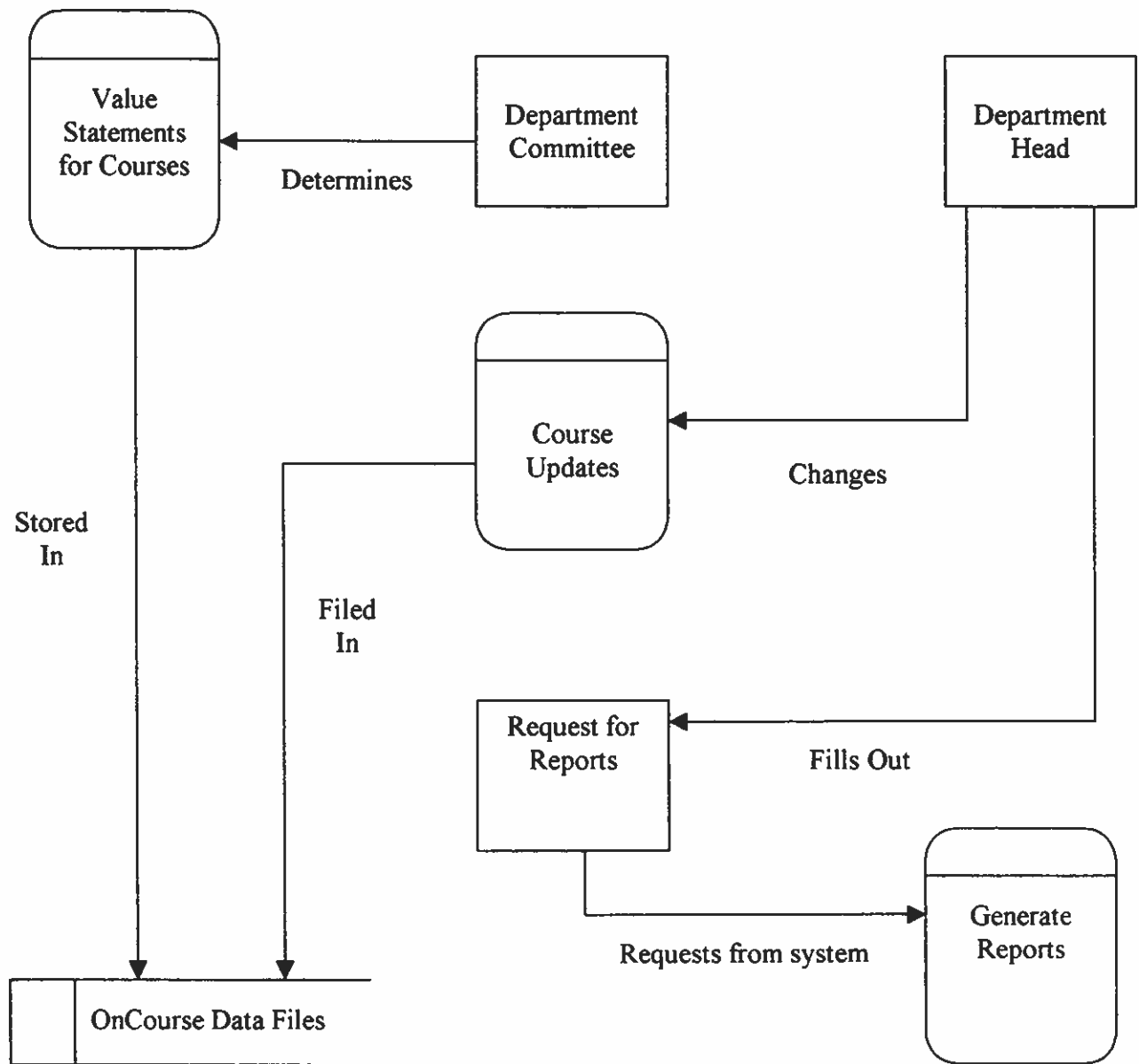
CONTEXT DIAGRAM



DECOMPOSITION DIAGRAM



SYSTEM DIAGRAM



EVENT LIST

Event Description	Trigger (Inputs)	Responses (Outputs)
Student Evaluates Course	Data-Entry program stores their data	Updates the database with the student's evaluation
Department Requests Analysis	Request made of the system to analyze the data for a course	Reports are made available for printing or viewing
Course Report Generated	Analysis is requested for all evaluations for a particular course	Statistics for the evaluations of that course are given
Courses Change	A course is added to the curriculum	The course database is updated
Departments Change	A department is added or information changes	The department database is updated
Professors Change	A professor is added or changed from a course	The professor database is updated
Value Statements are Changed	An authorized user (a department head) creates a new set of value statements	The value statements stored for that department are updated

CANDIDATE MATRIX

Characteristics	Candidate 1	Candidate 2	Candidate 3
Portion of System Computerized Brief description of that portion of the system that would be computerized in this candidate.	COTS software package designed to do basic statistical analysis of data.	After the data is manually inputted, the system would analyze the data and produce customized reports.	The system would have a computerized method of evaluating the courses and then would analyze the data and produce requested reports.
Benefits Brief description of the business benefits that would be realized for this candidate.	Would be a simplified and cost effective way to get the results.	Would be a customized program to give quick and informative reports that would be useful for the departments and the university.	Would eliminate the paperwork necessary for course evaluations and produce quick and informative reports that would be useful for the departments and the university.
Servers and Workstations A description of the servers and workstations needed to support this candidate.	Pentium Workstations that could load and run the software.	Pentium Workstations with Windows 95 or higher, Microsoft Access and Excel, and a network that could run the software.	Same as Candidate 2
Software Tools Needed Software tools needed to design and build the candidate (e. g., database management system, emulators, operating systems, languages, etc.). Not generally applicable if applications software packages are to be purchased.	No Tools Needed, Software Package Purchased	Microsoft Access and Microsoft Excel with a Visual Basic Front End.	Microsoft Access, Microsoft Excel, Microsoft Visual Basic and Java Programming.
Application Software A description of the software to be purchased, built, accessed, or some combination of these techniques.	Package Solution	Custom Solution	Custom Solution
Method of Data Processing Generally some combination of: on-line, batch, deferred batch, remote batch, and real-time.	Batch	Same as Candidate 1	Same as Candidate 1
Output Devices and Implications A description of output devices that would be used, special output requirements, (e.g. network, preprinted forms, etc.), and output considerations (e.g., timing constraints).	Preprinted Forms developed by the software that report the results of the statistical equations. Printed on HP5SI LAN Laser Printers.	Customized forms that would print desired reports on HP5SI LAN Laser Printers.	Same as Candidate 2
Input Devices and Implications A description of input methods to be used, input devices (e.g., keyboard, mouse, etc.), special input requirements, (e.g. new or revised forms from which data would be input), and input considerations (e.g., timing of actual inputs).	Keyboard, Mouse, Data Forms (Evaluations)	Same as Candidate 1	Keyboard, Mouse
Storage Devices and Implications Brief description of what data would be stored, what data would be accessed from existing stores, what storage media would be used, how much storage capacity would be needed, and how data would be organized.	Evaluation data would be stored on the PC hard drive that it was input to.	All evaluations, course, and department information would be stored in the main database files. The data would be kept on the server, but also distributed through the department workstations on backup systems for safety and security precautions.	Same as Candidate 2

FEASIBILITY MATRIX

Feasibility Criteria	Wt.	Candidate 1	Candidate 2	Candidate 3
Operational Feasibility Functionality. A description of to what degree the candidate would benefit the organization and how well the system would work. Political. A description of how well received this solution would be from both user management, user, and organization perspective.	60%	Would be a pre-purchased software package and could not be customized for their needs. Score: 60	Would support the evaluations, department information, and course information. Would analyze the data and produce custom reports. Score: 90	Same as Candidate 2 with an computerized evaluation process added. Score: 100
Technical Feasibility Technology. An assessment of the maturity, availability (or ability to acquire), and desirability of the computer technology needed to support this candidate. Expertise. An assessment to the technical expertise needed to develop, operate, and maintain the candidate system.	20%	Would be subject to upgrades from the pre-purchased software company. This Candidate would require a staff with the knowledge to install and maintain these upgrades. Score: 60	User friendly and designed for university faculty in mind. Would need minimal maintenance and could be upgraded upon request by Greenstone Consultants. Score: 100	Same as Candidate 2 only additional technical skill would be needed to maintain and upgrade the computerized evaluation system along with training and monitoring the usage. Score: 80
Economic Feasibility Cost to develop: Payback period (discounted): Net present value: Detailed calculations:	0%	Approximately \$ Approximately \$ Approximately \$ Score: unknown	Approximately \$ Approximately \$ Approximately \$ Score: unknown	Approximately \$ Approximately \$ Approximately \$ Score: unknown
Schedule Feasibility An assessment of how long the solution will take to design and implement.	20%	Less than 3 months Score: 100	Less than 8 months Score: 100	More than 1 year Score: 60
Ranking:	100%	68	94	88



SYSTEM PROPOSAL

I. Introduction

A. Purpose of the report

The purpose of this report is to review and select a candidate solution from those that have been proposed for the system that has been requested. This report will give information and explanation for the proposed candidates and will recommend the best choice, as chosen by Greenstone Consultants.

B. Background of the project leading to this report

An analysis of the systems request has been done, and Greenstone has determined that there are several approaches to fulfilling the request. Three of the candidates have been identified in the candidate matrix. The information is based on preliminary work done by Greenstone to evaluate the system and the needs of this project.

C. Scope of the project

This project involves a system that will automatically evaluate the results of student course evaluations. The scope of the project could be expanded to include an automated system of evaluations or could be limited to simply an analysis for the current system of evaluations.

D. Structure of the report

This report will give the recommended solution to the system request based on the work done by Greenstone Consultants and the operational, technical, economic, and schedule feasibility of the candidate based on the needs of the university.

II. Tools and techniques used

A. Solution Generated

Having reviewed the three candidates and their feasibility, Greenstone is recommending candidate 2 as the best solution to the operational, technical, economic, and schedule needs of the university. The system needs to be customized to meet the needs of the university and to produce customized reports upon request from the departments or the university. The system also needs to be easily maintained and used by the faculty of the university. Although some customized systems would meet these requirements, it is also important that the system be economically feasible and done in a reasonable amount of time.

B. Feasibility analysis (cost-benefit)

Candidate 2 will provide all the benefits of a system that is customized for the university's needs. It is the ideal candidate for a system that will analyze the student's course evaluations and produce reports that will benefit the professors, the departments, and the university. Candidate 2 will be created and supported by Greenstone Consulting. This system can be maintained and operated with very little technical knowledge necessary to the users. This candidate also complies with the economic feasibility of the university and can be completed and implemented within a reasonable amount of time.

III. Information systems requirements

This candidate would produce customized software that could be installed on any Pentium workstation that could support Microsoft Access and Excel or on a server and could be accessed from any of these workstations. The workstations could store

the software as well as the databases needed, and would require connection to a LAN printer to produce the customized reports.

IV. Alternative solutions and feasibility analysis

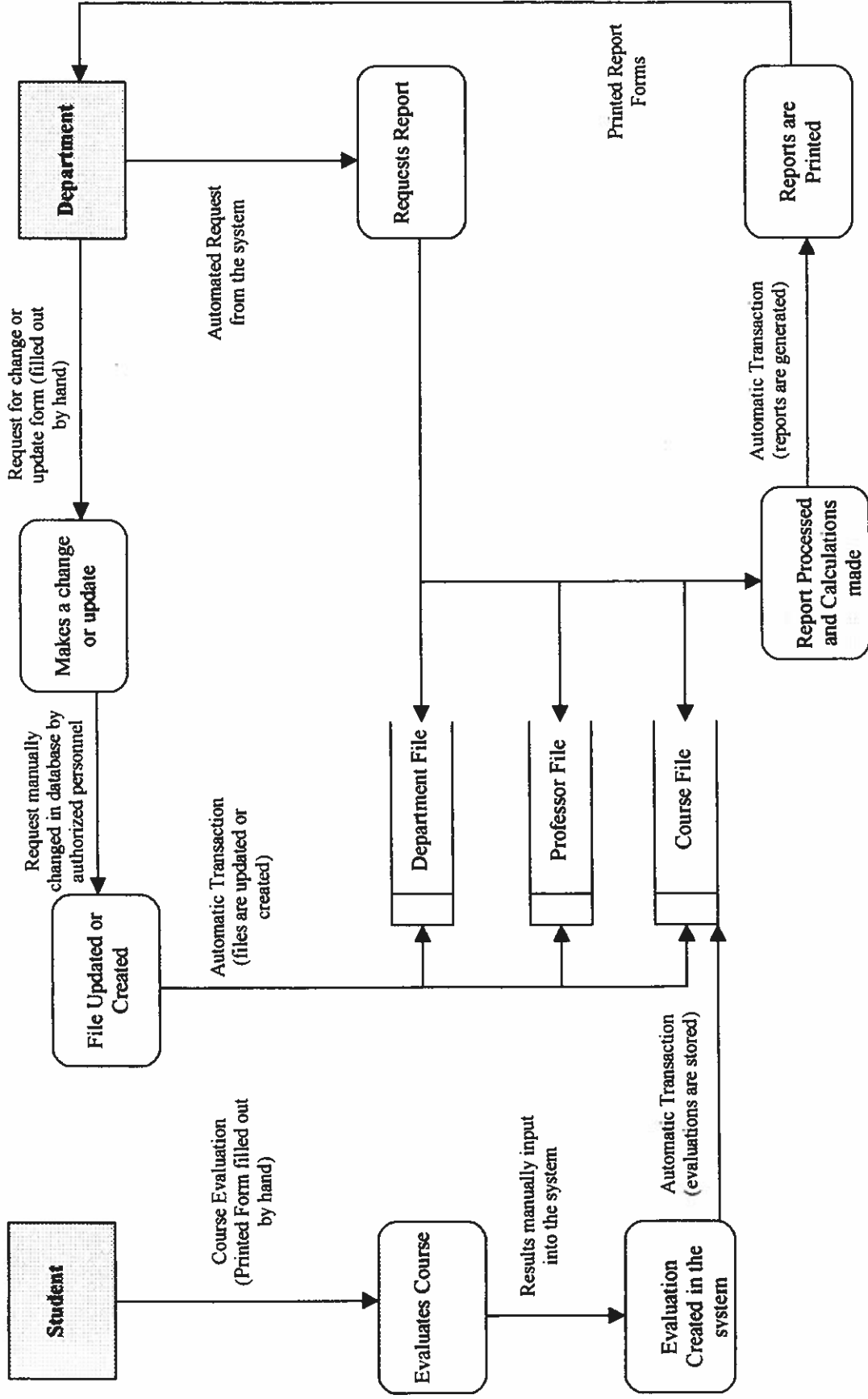
The alternative solutions are found in candidates 1 and 3. These candidates propose systems that are either below the operational desires of the university or above the economic and schedule feasibility. Candidate 1 proposes a pre-packaged statistical system that could be purchased and used to run basic statistical calculations. This system would not produce the customized results that are desired, and would not be flexible to changes and upgrades from Greenstone. This system would be cost effective and timely, but the benefits would be so few that the cost would not outweigh it.

Candidate 3 would be an extensive system that would meet all the needs requested system. It would require advanced programming which would make it an expensive and time consuming project that somewhat outweighs the benefits that would come from this advanced system. It would also require more technical training and maintenance because there would be a wide scope of users for the computerized evaluation system.

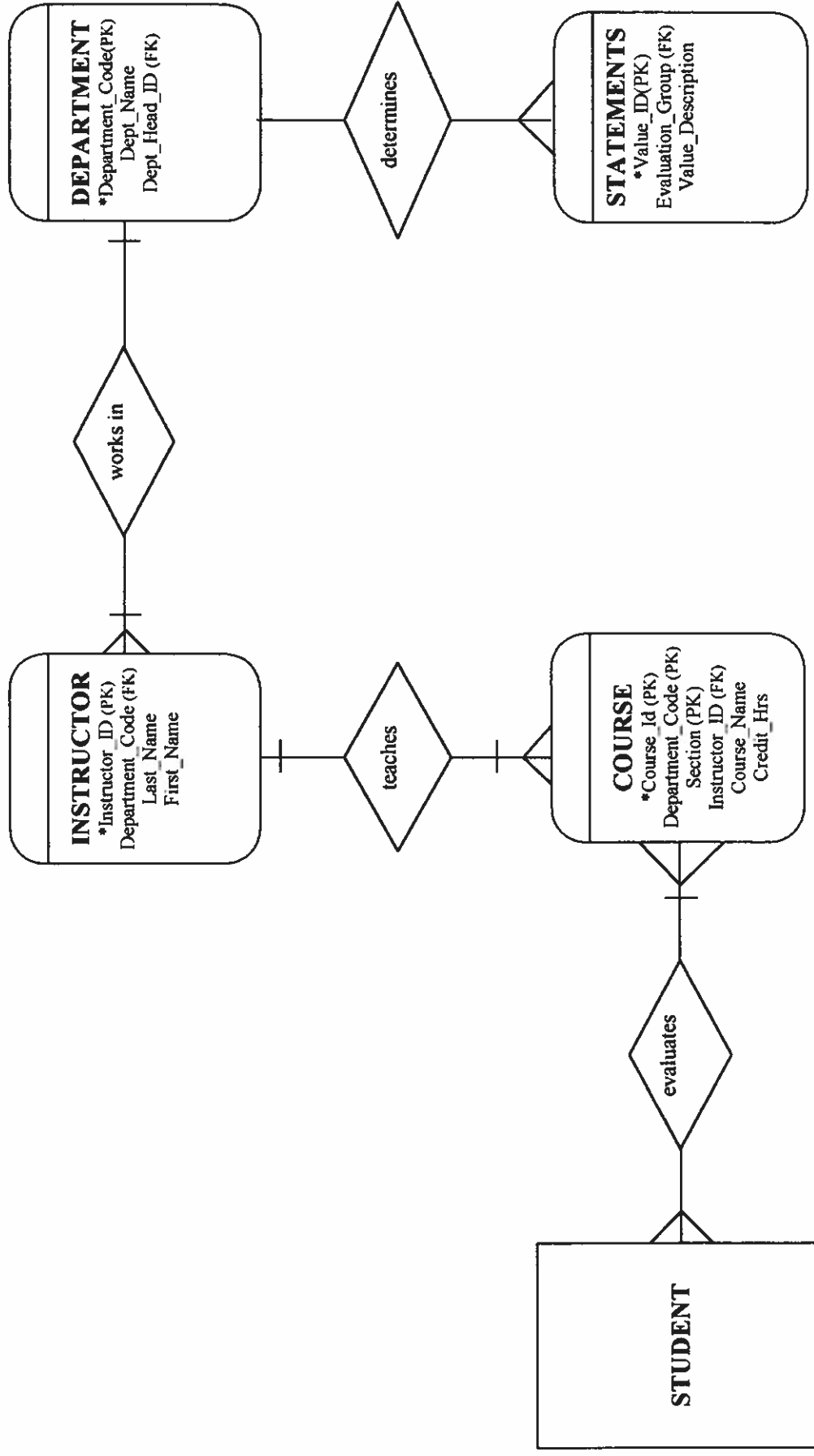
V. Recommendations

Based on the analysis of the three candidates, Greenstone Consulting is recommending candidate 2 as the best system proposal to meet the needs of the university. Candidate 2 is the best combination of operational, technical, economic and schedule feasibility amongst the three candidates. It is a customized system that can be easily used and maintained and will provide the results that are desired. Candidate 2 is the top choice by Greenstone to meet the needs of the university.

PHYSICAL DATA FLOW DIAGRAM



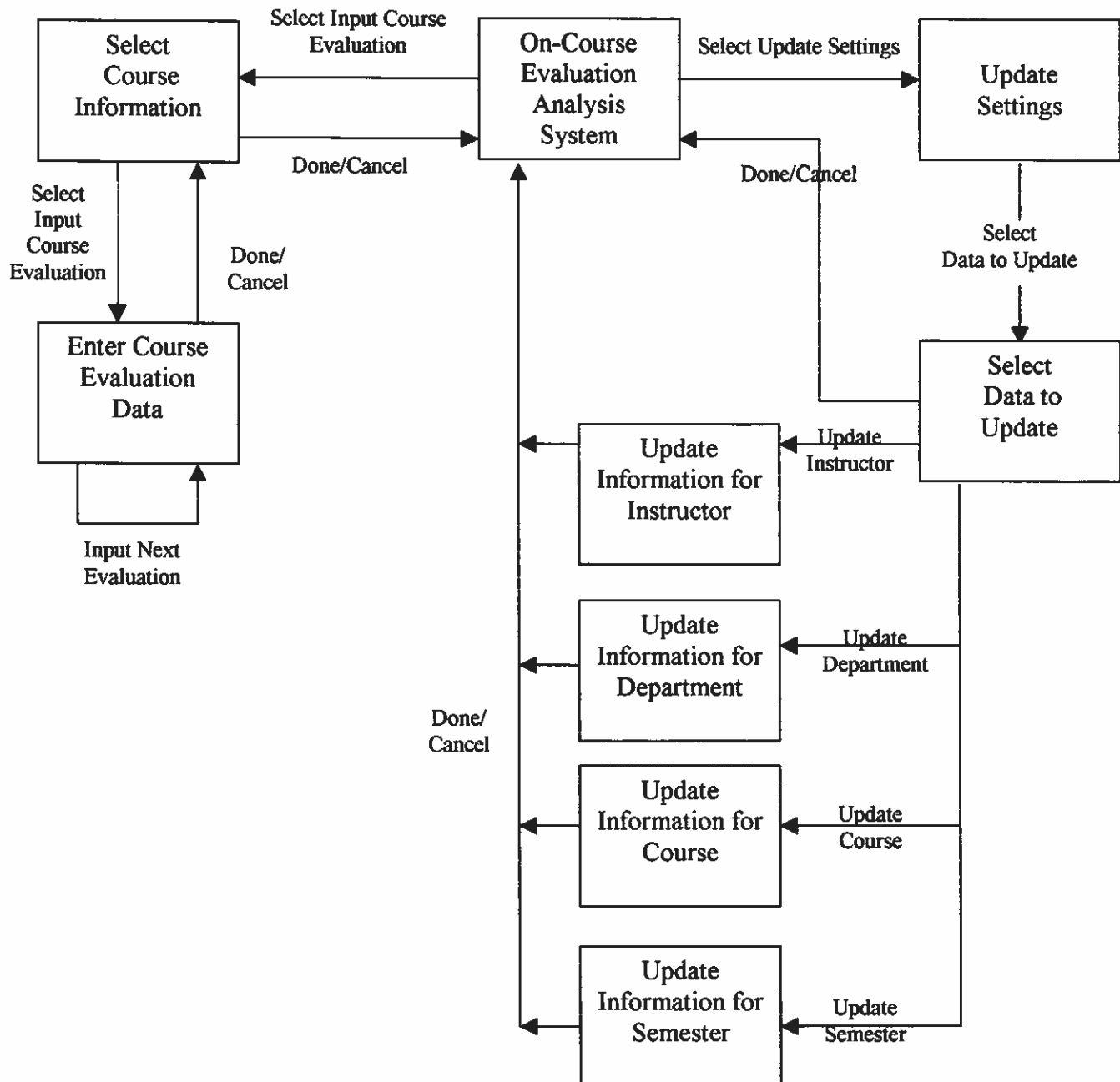
Normalized Relations Diagram



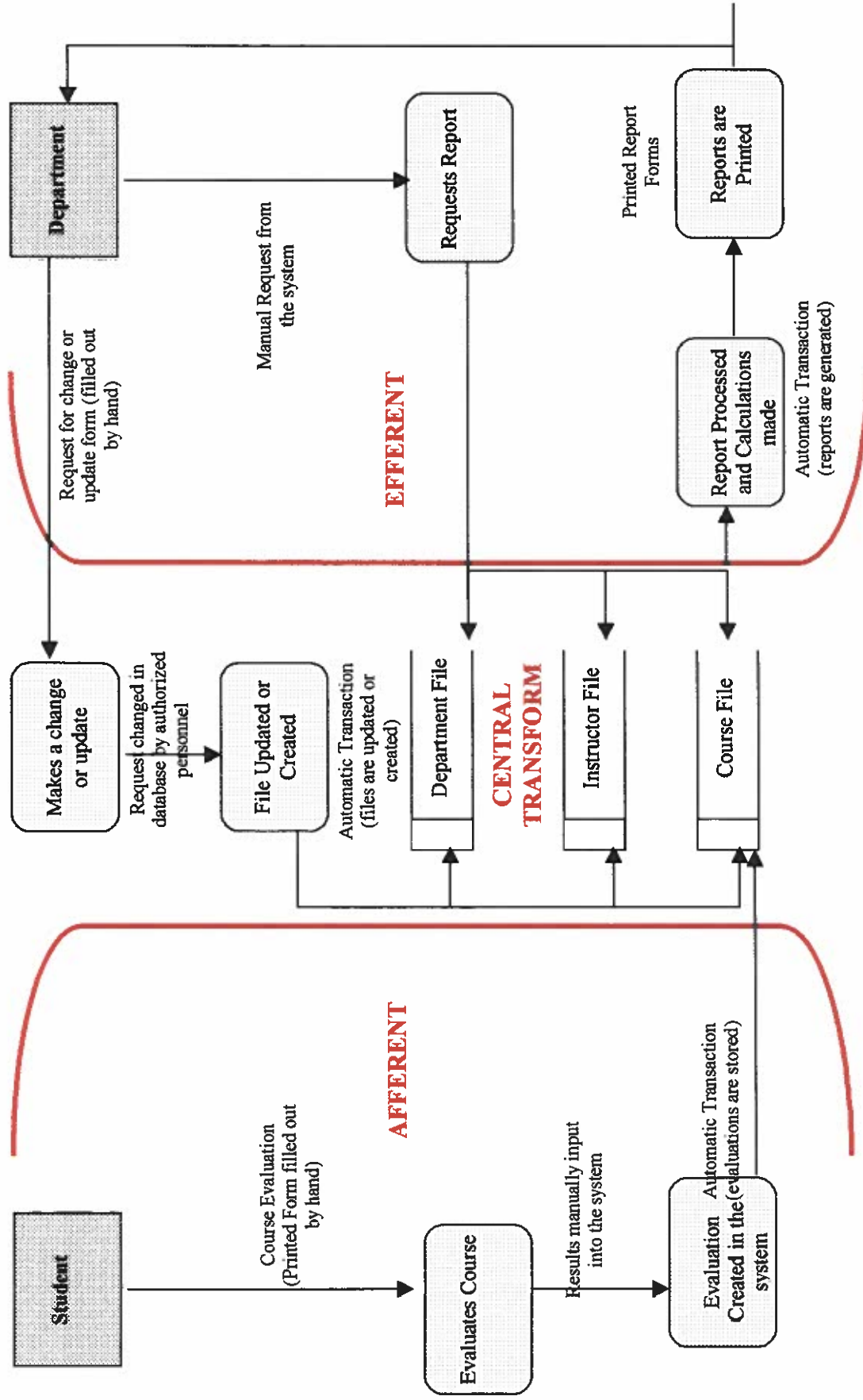
COMPLETE TABLE DEFINITIONS

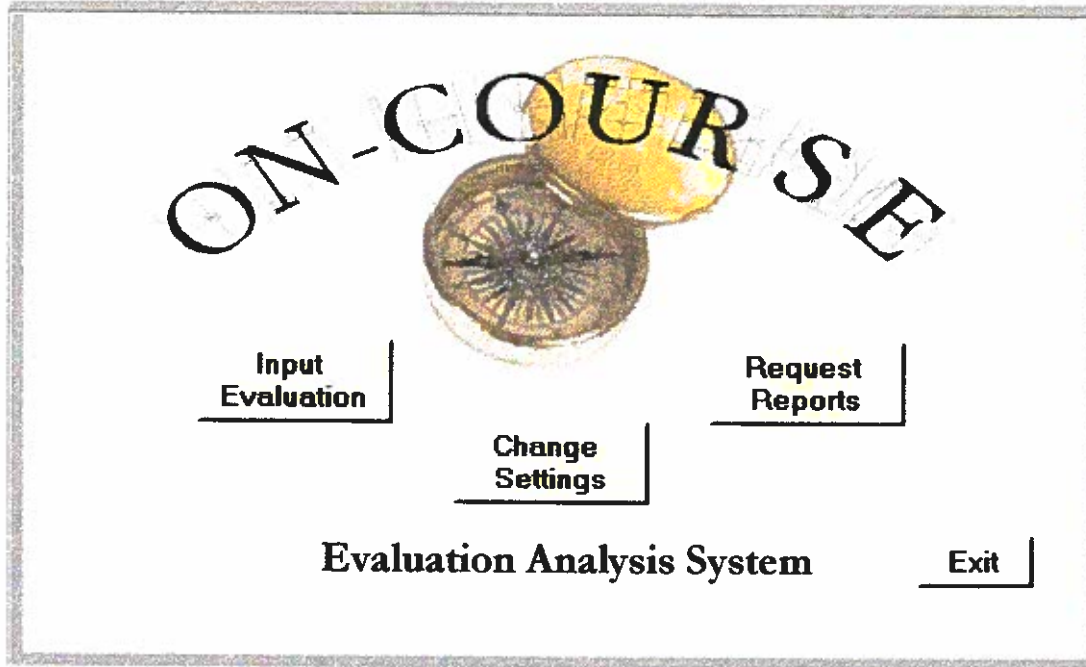
Table Name	Field Name	Type	Size	Description	Mask
Course	Department_Code (PK)	VARCHAR	4		AAAA
	Course_Id (PK)	CHAR	3		###
	Section (PK)	CHAR	2		##
	Semester (FK)	VARCHAR	10		
	Instructor_Id (FK)	CHAR	3		###
	Course_Name	VARCHAR	30		
	Credit_Hrs	CHAR	1		#
Instructor	Instructor_ID (PK)	CHAR	3		###
	Department_Code (FK)	VARCHAR	4		AAAA
	Last_Name	VARCHAR	20		
	First_Name	VARCHAR	20		
Department	Department_Code (PK)	VARCHAR	4		AAAA
	Dept_Head_Id (FK)	CHAR	3		###
	Dept_Name	VARCHAR	20		
Value Statements	Value_Id (PK)	CHAR	2		##
	Evaluation_Group	VARCHAR	4		AAAA
	Value_Description	VARCHAR	50		

STATE TRANSITION DIAGRAM



PARTITIONED DATA FLOW DIAGRAM





Main Form

frmMain.frm

This is the main form of the On-Course Program. This form allows the user to select one of 3 main functions of the program.

- **Input Evaluation**
- **Change Settings**
- **Request Reports**

When the user selects “Input Evaluation”, it will bring up the screen to Select Course Information. This will be the information about the course that is being evaluated.

When the user selects “Change Settings”, it will bring up the screen to Change Settings, giving them the option to update several tables that are located in the On-Course database.

At the current time, “Request Reports” is not functioning since the available reports are not directly connected to the Visual Basic program. On-Course can be modified in the future to include this function.

“Exit” will terminate the program and close all open screens.

Main Form

Code for frmMain.frm

```
Private Sub cmdExit_Click()
```

```
' Closes all open screens and terminates the program  
Unload Me
```

```
End Sub
```

```
Private Sub cmdInputEvaluation_Click()
```

```
'Opens frmInputEvaluation1 which allows the user to select the course information  
Load frmInputEvaluation1  
frmInputEvaluation1.Show  
frmInputEvaluation1.Refresh
```

```
End Sub
```

```
Private Sub cmdRequestReports_Click()
```

```
'Opens frmReports which allows the user to select report options  
Load frmReports  
frmReports.Show
```

```
End Sub
```

```
Private Sub cmdSettings_Click()
```

```
' Opens frmSetting which allows the user to add data to the OnCourse data tables  
Load frmSettings  
frmSettings.Show
```

```
End Sub
```

Select Course Information

Instructor:

Semester:

Department Code:

Course Number:

Section:

 Main Menu



Select Course Information

frmInputEvaluation1.frm

This form is pulled up when the user selects “Input Evaluation” from the Main form. This screen consists of 4 combo boxes and a text box.

When the user selects an Instructor_ID from the Instructor combo box, the form automatically fills in the department code for that particular instructor. All subsequent combo boxes are locked until the previous selection is made. For instance, the user is unable to choose a section until a course number has been selected.

Each combo box on this screen is dependant on the previous boxes in order to narrow down the selection. When a semester is chosen, the course numbers listed are only those taught by the selected instructor during the selected semester. The section is again narrowed down to only those of the selected course number.

If the user wishes to change the data that has been selected, re-selecting an Instructor_ID from the Instructor box will update the department code and clear out all previous selections.

When all course information has been entered, the user may select “Input Evaluation” to begin entering the students’ evaluations.

“Main Menu” will close this screen and return back to the Main form.

Select Course Information Form

Code for frmInputEvaluation1.frm

Public Sub Fill(typeFill As String)

'Sets up cases that will be called to fill each combo box at the appropriate time

Select Case typeFill

Dim cnn As New ADODB.Connection 'establishes ADODB connection to the database

Dim rs As New ADODB.Recordset

Dim cnstr As String

Dim sqlStr As String

Dim returnVal As Integer

Dim instrCode As String

SAMPLE CODE FOR CASE "INSTRUCTOR" -- repeated and modified for all other cases (Department, Semester, Course, Section)

'FILL INSTRUCTOR COMBO BOX

Case "Instructor"

returnVal = 0

sqlStr = "Select * from Instructor"

cnstr = "Driver={Microsoft Access Driver (*.mdb)};Dbq=c:\OnCourse\oncourse.mdb;Uid=;Pwd="

With cnn

.Open cnstr 'opens connection to On-Course database

End With

With rs

.CursorType = adOpenForwardOnly

.CursorLocation = 3

.LockType = adLockOptimistic

.Open sqlStr, cnn

End With

'Adds the records from Instructor table to the combo box

If rs.RecordCount > 0 Then

Do Until rs.EOF

frmInputEvaluation1.cmbInstructor.AddItem (rs.Fields("Instructor_ID").Value)

rs.MoveNext

Loop

Else

yn = MsgBox("There was no data entered in the Instructor table.", vbCritical + vbOKOnly, "Critical

Error")

End If

rs.Close

cnn.Close

End Select

End Sub

Private Sub cmbCourse_Click()

'Enables and fills Section combo box once course has been selected

frmInputEvaluation1.cmbSection.Enabled = True

frmInputEvaluation1.cmbSection.Clear

Call Fill("Section")

End Sub

Code for frmInputEvaluation1.frm (cont...)

```
Private Sub cmbInstructor_Click()  
'Enables and fills Semester and Department once Instructor_ID had been selected  
frmInputEvaluation1.cmbSemester.Enabled = True  
frmInputEvaluation1.cmbSemester.Clear  
frmInputEvaluation1.cmbCourse.Clear  
frmInputEvaluation1.cmbSection.Clear  
Call Fill("Semester")  
Call Fill("Dept")  
End Sub
```

```
Private Sub cmbSection_Click()  
'Enables the user to select "Input Evaluation" once all fields have been selected  
frmInputEvaluation1.cmdInputEvaluation.Enabled = True  
End Sub
```

```
Private Sub cmbSemester_Click()  
'Enables and fills Course combo box once the semester has been selected  
frmInputEvaluation1.cmbCourse.Enabled = True  
frmInputEvaluation1.cmbCourse.Clear  
frmInputEvaluation1.cmbSection.Clear  
Call Fill("Course")  
End Sub
```

```
Private Sub Form_Activate()  
'Enables and fills the Instructor combo box when the form is initialized  
Dim cnn As New ADODB.Connection  
Dim rs As New ADODB.Recordset  
Dim cnstr As String  
Dim sqlStr As String  
Dim returnVal As Integer  
    Call Fill("Instructor")  
End Sub
```

```
Private Sub cmdInputEvaluation_Click()  
'Loads the frmEval that will allow the user to input the evaluations for the selected course  
Load frmEval  
frmEval.Show  
End Sub
```

```
Private Sub cmdReturn_Click()  
'Returns to the Main Menu  
Unload Me  
Load frmMain  
frmMain.Show  
End Sub
```

ON-COURSE

NEUTRAL or UNIMPORTANT

LEAST like course				NEUTRAL or UNIMPORTANT					MOST like course	
lstBox1	lstBox2	lstBox3	lstBox4	lstBox5	lstBox6	lstBox7	lstBox8	lstBox9		
1									1	
	3						3			
		4								
Student			5		5					
Instructor				6						
Course				lstValue						
Semester										

Input Evaluation



Main Menu



Input Next Evaluation

Evaluation Form

frmEval.frm

This form is brought up from the Select Course Information Screen when the user selects "Input Evaluation". The course information that the user has selected will automatically appear in the text boxes on the bottom left of the screen. The user **MUST** enter a student number (beginning with 1) for each student in a particular course before the data from this screen can be saved into the On-Course database.

The values contained in the list box at the bottom center of the screen correspond to the value statements 1 – 32 that were given to evaluate the course. The students have placed each value in one of the list boxes 1-9. List box 1 being LEAST like the course and list box 9 being MOST like the course. The values from the bottom list box must be dragged and dropped into the appropriate list box corresponding with the students' evaluations.

When a value is moved from the bottom box into one of the top list boxes, the value will be removed as a selection from the list. This is to ensure that all values are used once and only once. When all the values have been dragged into the appropriate list box and the student number has been entered, the user must click "Input Next Evaluation" in order to save the data into the On-Course database and bring up a clear screen for the next evaluation.

Input Evaluation Form

Code for frmEval.frm

```
Private Sub cmdInput_Click()
'Inputs the student evaluations into the data table of the On-Course database
Dim cnn As New ADODB.Connection
Dim rs As New ADODB.Recordset
Dim cnstr As String
Dim sqlStr As String
Dim returnVal As Integer
'FILL DATA IN DB
cnstr = "Driver={Microsoft Access Driver (*.mdb)};Dbq=c:\OnCourse\oncourse.mdb;Uid=;Pwd="
With cnn
    .Open cnstr      'establishes a connection to On-Course database
End With
returnVal = 0
sqlStr = "Select * from data where student='XXXXXXXXXX'"
With rs
    .CursorType = adOpenForwardOnly
    .CursorLocation = adUseClient
    .LockType = adLockOptimistic
    .Open sqlStr, cnn
End With
'Verifies that the user has entered the student number
If Trim(frmEval.txtStudent.Text) = "" Then
    yn = MsgBox("You must enter a Student number.", vbOKOnly, "Data Error")
    Exit Sub
End If
If rs.RecordCount = 0 Then
    This loops through listbox 1 and stores the info from the box
    ***SAMPLE CODE FOR STORING DATA FROM THE LISTBOX – repeated and modified for
    lstBox 3 through 9***
    For I = 0 To frmEval.lstBox1.ListCount - 1
        rs.AddNew
        rs.Fields("code").Value = Trim(frmEval.lstBox1.List(I))
        rs.Fields("box_number").Value = "1"
        rs.Fields("row_number").Value = I
        rs.Fields("course").Value = Trim(frmEval.txtCourse.Text)
        rs.Fields("student").Value = Trim(frmEval.txtStudent.Text)
    Next I
    For I = 0 To frmEval.lstBox2.ListCount - 1
        rs.AddNew
        rs.Fields("code").Value = Trim(frmEval.lstBox2.List(I))
        rs.Fields("box_number").Value = "2"
        rs.Fields("row_number").Value = I
        rs.Fields("course").Value = Trim(frmEval.txtCourse.Text)
        rs.Fields("student").Value = Trim(frmEval.txtStudent.Text)
    Next I
    rs.Update
End If
rs.Close
```

Code for frmEval.frm (cont...)

```
cnn.Close
Unload Me
Load frmEval
frmEval.Show
End Sub
```

```
Private Sub Form_Activate()
Dim cnn As New ADODB.Connection
Dim rs As New ADODB.Recordset
Dim cnstr As String
Dim sqlStr As String
Dim returnVal As Integer
'Keeps the course information stored until the "Select Course Information" screen is changed
txtInstructor.Text = Trim(frmInputEvaluation1.cmbInstructor.Text)
txtCourse.Text = Trim(frmInputEvaluation1.txtDepartment.Text) + " " +
Trim(frmInputEvaluation1.cmbCourse.Text) + " " + Trim(frmInputEvaluation1.cmbSection.Text)
txtSemester.Text = Trim(frmInputEvaluation1.cmbSemester.Text)
returnVal = 0
sqlStr = "Select * from Statements"
cnstr = "Driver={Microsoft Access Driver (*.mdb)};Dbq=c:\OnCourse\oncourse.mdb;Uid=;Pwd="
With cnn
    .Open cnstr
End With
With rs
    .CursorType = 2
    .CursorLocation = 3
    .LockType = 3
    .Open sqlStr, cnn
End With
'Fills the lstValue list box
If rs.RecordCount > 0 Then
    Do Until rs.EOF
        frmEval.lstValueSt.AddItem (rs.Fields("Value_Id").Value)
        rs.MoveNext
    Loop
Else
    yn = MsgBox("There was no data entered in the Statements table.", vbCritical + vbOKOnly, "Critical
Error")
End If
rs.Close
cnn.Close
End Sub
```

SAMPLE CODE FOR DRAG & DROP – modified and repeated for all 9 list boxes

'Allows the user to drag & drop the values into the appropriate list boxes

```
Private Sub lstBox1_DragOver(Source As Control, X As Single, Y As Single, State As Integer)
Dim curVal As String
Dim I As Integer
Dim curValIndex As Integer
curValIndex = Source.ListIndex
curVal = Trim(Source.List(Source.ListIndex))
```

Code for frmEval.frm (cont...)

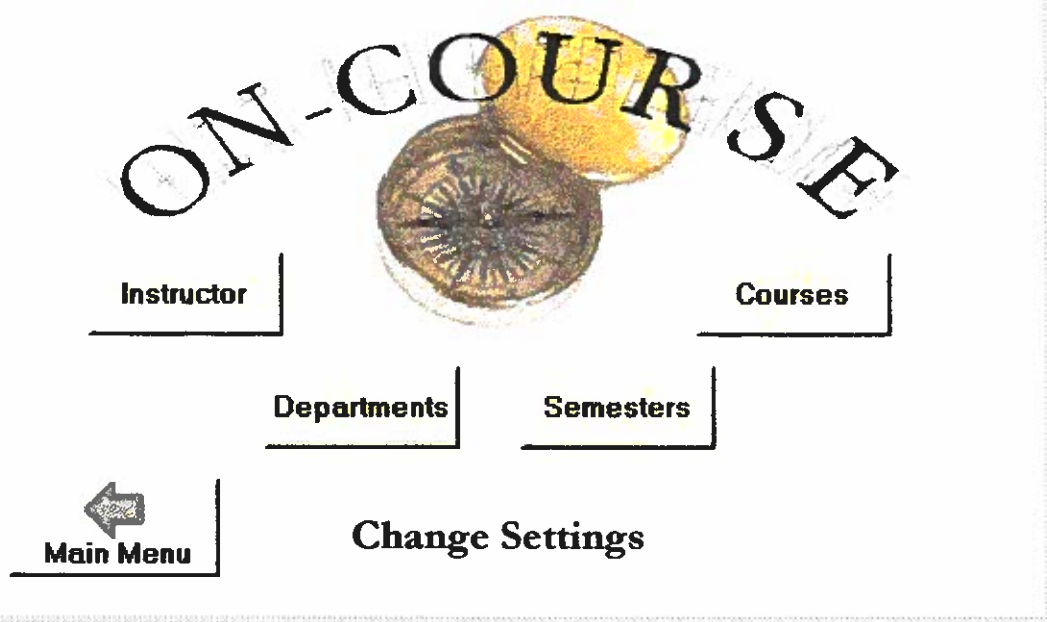
```
If Trim(curVal) = "" Then
    Exit Sub
End If
For I = 0 To frmEval.lstBox1.ListCount - 1 ' Loop through list.
    If Trim(curVal) = frmEval.lstBox1.List(I) Then
        Exit Sub
    End If
Next I
If frmEval.lstBox1.ListCount < 1 Then
    If State = vbEnter Then
        frmEval.lstBox1.AddItem (curVal)
        Source.RemoveItem (curValIndex)
    End If
End If
Source.Drag (0)
End Sub
```

'Picks up the value that the mouse is clicked on

```
Private Sub lstBox1_MouseDown(Button As Integer, Shift As Integer, X As Single, Y As Single)
If frmEval.lstBox1.List(frmEval.lstBox1.ListIndex) = "" Then
    frmEval.lstBox1.Drag (0)
Else
    frmEval.lstBox1.DragMode = 0
    frmEval.lstBox1.Drag (1)
End If
End Sub
```

```
Private Sub lstBox1_MouseUp(Button As Integer, Shift As Integer, X As Single, Y As Single)
    frmEval.lstBox1.DragMode = 0
    frmEval.lstBox1.Drag (2)
End Sub
```

```
Private Sub lstValueSt_DragOver(Source As Control, X As Single, Y As Single, State As Integer)
    Dim curVal As String
    Dim I As Integer
    Dim curValIndex As Integer
    curValIndex = Source.ListIndex
    curVal = Trim(Source.List(Source.ListIndex))
    For I = 0 To frmEval.lstValueSt.ListCount - 1 ' Loop through list.
        If Trim(curVal) = frmEval.lstValueSt.List(I) Then
            Exit Sub
        End If
    Next I
    'removes the value from the lstValue once it has been put into another list box
    If frmEval.lstValueSt.ListCount < 1 Then
        If State = vbEnter Then
            frmEval.lstValueSt.AddItem (curVal)
            Source.RemoveItem (curValIndex)
        End If
    End If
    Source.Drag (0)
End Sub
```



Change Settings

frmSettings.frm

When the user selects “Change Settings” from the Main form, this screen appears. The feature to change settings allows the user to automatically add data to the On-Course database. Each selection is linked to an appropriate form for entering new data, thus the code for this form is fairly minimal.

“Change Setting” offers the following tables to be updated:

- Instructor
- Departments
- Semesters
- Courses

Currently these options are available for automatic updates as they are the most dynamic tables. There are other tables in the On-Course database that may be edited as needed through the database itself such as semesters, value statements, and evaluation groups.



Add Course

Semester*		Instructor_ID*	
Department Code*		Credit Hours	
Course_ID		*If new Instructor, Department, or Semester, those tables must first be updated	
Section			
Main Menu		Add Course	

Add Instructor
frmAddInstr.frm

Add Departments
frmAddDept.frm

Add Semester
frmAddSemester.frm

Add Course
frmAddCourse.frm

(Add Course (frmAddCourse.frm) is shown above)

When a user selects an option from the “Change Settings” screen, a form is brought up that will allow the user to enter data into a table of the On-Course database.

Some of the selections have drop down combo boxes where information must be chosen from data already in a database such as department code or instructor ID. If a course is being added with a new instructor, semester, or department; the instructor ID, department code, and semester must be entered in their own database prior to creating a course with this information.

Each combo or text box on the form should be filled out to ensure that the database has all the required information to process a course evaluation.

Once the data has been entered, the “Add” button will then save the information into the appropriate table of the On-Course database. The user can then enter another update or click “Main Menu” to return to the Main Form.

Change Setting Forms

Sample Code for frmAddInstr.frm

```
Private Sub cmdAddInstructor_Click()
Dim cnn As New ADODB.Connection
Dim rs As New ADODB.Recordset
Dim cnstr As String
Dim sqlStr As String
Dim returnVal As Integer

'FILL INSTRUCTOR TABLE IN DB
cnstr = "Driver={Microsoft Access Driver (*.mdb)};Dbq=c:\OnCourse\oncourse.mdb;Uid=;Pwd="
With cnn
.Open cnstr
End With

returnVal = 0
sqlStr = "Select * from Instructor where Instructor_ID='XXXXXXXXXX'"

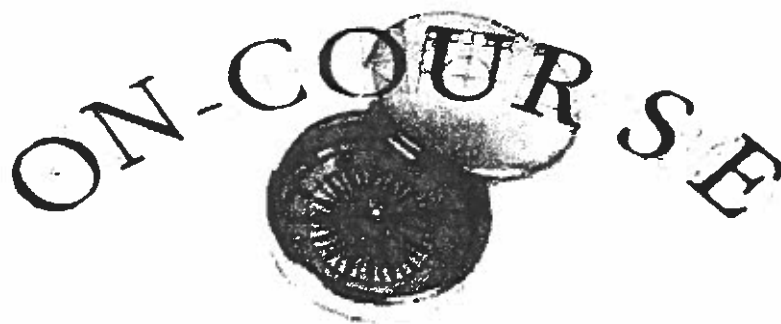
With rs
.CursorType = adOpenForwardOnly
.CursorLocation = adUseClient
.LockType = adLockOptimistic
.Open sqlStr, cnn
End With

If rs.RecordCount = 0 Then
This stores the info into the Instructor database
rs.AddNew
rs.Fields("Instructor_ID").Value = Trim(frmAddInstr.txtInstructorID.Text)
rs.Fields("First_Name").Value = Trim(frmAddInstr.txtFirstName.Text)
rs.Fields("Last_Name").Value = Trim(frmAddInstr.txtLastName.Text)
rs.Fields("Department_Code").Value = Trim(frmAddInstr.cmbDept.Text)
rs.Update
End If

rs.Close
cnn.Close

Unload Me
Load frmAddInstr
frmAddInstr.Show

End Sub
```



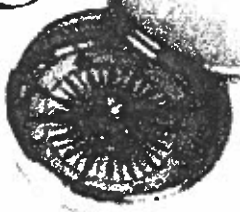
Microsoft Access

oncourse.mdb

Tables

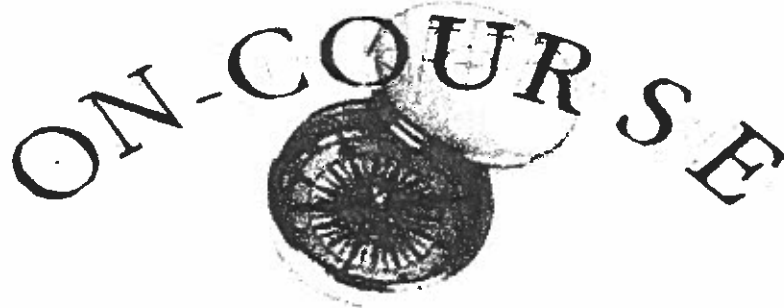
CourseList	Stores all course information
	Semester, Department_Code, Course_ID, Section, Instructor_ID, Credit_Hrs
Data	Stores all student evaluation data automatically from frmEval.frm
	Course, Student, Code, Box_Number, Row_Number
Data Template	Makes sure there is room for at least 30 student evaluations per class – can be adjusted
	CodeN, BoxN, StudentN, Course
Department	Stores all department information
	Department_Code, Department_Name, Dept_Head_ID
DesiredDataResults	Stores the ideal evaluation results as determined by the department or university
	Code, Box_Number
EvaluationGroup	Stores which departments belong to which evaluation groups. Currently all belong to 1 but can be changed in the future if a variety of value statements are desired
	Evaluation_Group, Department_Code
Instructor	Stores all instructor information
	Instructor_ID, First_Name, Last_Name, Department_Code
Semester	Stores all semester information
	Semester
Statements	Stores all value statements
	Evaluation_Group, Value_ID, Value_Description

ON-COURSE



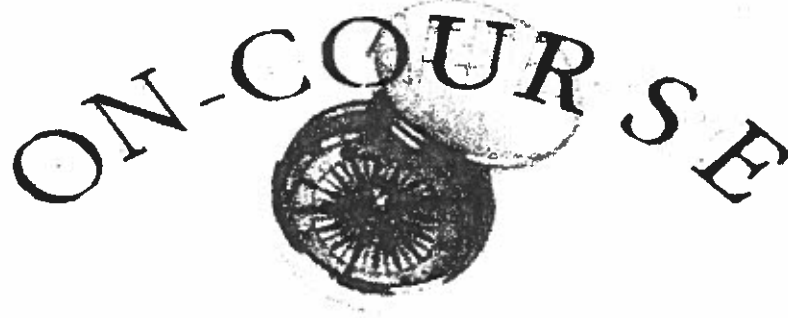
Queries

ActualVsDesiredQuery	Pulls information from DesiredData table, EvalReportQuery#1, and InstructorQuery to give the actual mean for each statement compared to the desired value for each course
	Course, Code, Mean, DesiredValue, Offset, Value_Description, Instructor_Name
AverageByItem&Course	Lists each value statement and the actual mean for each statement for all classes evaluated
	Code, Value_Description, & Mean for each class
DataConversion	Converts data from text fields entered from VB to number fields that can be used for calculations
	StudentN, BoxN, CodeN, Course, Instructor_Name
DataQuery	Union between DataConversionQuery and DataTemplate Table
	Code, Box_Number, Student, Course
EvalReportQuery#1	Lists where each student put each value statement for a particular class and the mean for each statement
	Course, Instructor_Name, Code, Student (1-n), Mean, Value_Description, Evaluation_Group
InstructorQuery	Combines first & last name for Instructor to make Instructor_Name & Department_Code, Course_ID, and Section to make Course
	Instructor_Name, Course
MeansByCourseAndItem	Gives average box_number value for each item from a particular course
	Course, Code, AvgofBox_Number
Names	Combines instructor first& last name to make full_Name and links it to Instructor_ID
	Full_Name, Instructor_ID
StudentCrosstab	Lists course evaluated, and value each student gave for each value statement
	Course, Code, Student(1-n)



Reports

ActualVsDesiredReport	Lists the value (1-32), the mean for each value, the desired value, the difference between the actual and desired, and the value statements for each course
CourseEvaluationSummary-ByItem	Lists the value (1-32), the students (1-n), where each student placed each value in the evaluation, the mean for each value, and the value statements for each course – Sorted by Value Statements
CourseEvaluationSummary-ByMean	Lists the value (1-32), the students (1-n), where each student placed each value in the evaluation, the mean for each value, and the value statements for each course – Sorted by Mean



Macros

WriteSummarytoExcel	Writes the AverageByItem&Course Query out to Microsoft Excel, saving it as an .xls
WriteTabletoExcel	Writes the ActualVsDesiredQuery out to Microsoft Excel, saving it as an .xls

OnCourse - ActualVsDesired Offset Values

ART 433 01

Aaron Baldwin

Item	Mean	DesiredValue	Offset	Item Description
31	6.33	1	5.33	Teacher is intimidating, condescending, uncompromising
26	6.67	2	4.67	Instructor speaks in monotone voice
19	7.00	3	4.00	Lectures too much
11	5.00	2	3.00	Class is slow, tedious, boring
25	6.33	4	2.33	Enjoyable and funny
30	7.33	5	2.33	Class participation is encouraged
22	5.00	3	2.00	Too much time spent on simple issues
15	4.67	3	1.67	Tests are easy
28	5.67	4	1.67	Workload is heavy
17	5.33	4	1.33	Tests/assignments are returned in a reasonable time
18	4.00	3	1.00	Teaches from the book
1	6.00	5	1.00	Tests related to what is covered in class
32	7.00	6	1.00	Instructor applies subject to the real world
29	7.00	6	1.00	Teacher is well prepared for each class
14	5.00	4	1.00	Tests are mostly essay and short answer
12	5.67	5	0.67	Instructor is punctual and attends class regularly
3	2.67	2	0.67	Talks over your head
13	4.33	4	0.33	Must take good note to be successful
7	5.00	5	0.00	Professor is always available to help
16	6.33	7	-0.67	I learned a lot
5	4.00	5	-1.00	Instructor comments on written work were useful
2	4.00	5	-1.00	Must do homework and be prepared for class
21	3.67	6	-2.33	Professor is fair
27	4.33	7	-2.67	Thought provoking
6	4.00	7	-3.00	Instructor expectations are clear and consistent
24	3.00	6	-3.00	Challenging and demanding
10	3.00	6	-3.00	Clear, well structured presentations
9	4.67	8	-3.33	Teacher sets high academic standards
20	4.67	8	-3.33	Stimulates interest in the subject
4	3.33	7	-3.67	Professor knows subject matter
23	5.33	9	-3.67	Profesor treats students with respect
8	3.67	8	-4.33	Instructor explains concepts well

OnCourse - Course Evaluation Summary - By Item

Course: COIN 325 91

Instructor: James Roberts

-- Student --

Item	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	Mean	Value	Description
1	5	5	5	8	5	8																					6.00	Tests related to what is covered in class	
2	1	4	5	4	6	6																					4.33	Must do homework and be prepared for class	
3	3	5	4	3	3	4																					3.67	Talks over your head	
4	5	5	4	5	8	5																					5.33	Professor knows subject matter	
5	4	7	9	6	9	5																					6.67	Instructor comments on written work were useful	
6	6	2	1	1	8	2																					3.33	Instructor expectations are clear and consistent	
7	4	9	3	4	5	3																					4.67	Professor is always available to help	
8	5	3	3	7	7	6																					5.17	Instructor explains concepts well	
9	4	6	3	8	2	3																					4.33	Teacher sets high academic standards	
10	6	4	5	8	5	4																					5.33	Clear, well structured presentations	
11	4	3	7	6	4	1																					4.17	Class is slow, tedious, boring	
12	4	5	2	4	1	7																					3.83	Instructor is punctual and attends class regularly	
13	2	1	2	5	5	7																					3.67	Must take good note to be successful	
14	5	6	6	3	4	5																					4.83	Tests are mostly essay and short answer	
15	2	4	7	5	4	8																					5.00	Tests are easy	
16	3	5	6	2	6	9																					5.17	I learned a lot	
17	6	3	8	3	3	6																					4.83	Tests/assignments are returned in a reasonable time	
18	8	3	6	5	4	3																					4.83	Teaches from the book	
19	7	2	7	7	2	3																					4.67	Lectures too much	
20	7	4	4	6	7	7																					5.83	Stimulates interest in the subject	
21	6	4	8	5	7	4																					5.67	Professor is fair	
22	9	5	4	2	2	2																					4.00	Too much time spent on simple issues	
23	2	2	2	2	6	4																					3.00	Professor treats students with respect	
24	8	6	5	3	3	2																					4.50	Challenging and demanding	
25	5	8	4	4	8	5																					5.67	Enjoyable and funny	
26	3	7	3	7	5	8																					5.50	Instructor speaks in monotone voice	
27	5	6	5	4	4	6																					5.00	Thought provoking	
28	6	8	5	6	6	7																					6.33	Workload is heavy	
29	7	6	7	7	5	5																					6.17	Teacher is well prepared for each class	
30	3	7	6	5	3	6																					5.00	Class participation is encouraged	
31	7	8	8	6	7	5																					6.83	Teacher is intimidating, condescending, uncompromising	
32	8	7	6	9	6	4																					6.67	Instructor applies subject to the real world	

OnCourse - Course Evaluation Summary - By Mean

Course: COIN 325 91

Instructor:

James Roberts

-- Student --

Item	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Mean	Value_Description
31	7	8	8	6	7	5																			6.83	Teacher is intimidating, condescending, uncompromising
5	4	7	9	6	9	5																			6.67	Instructor comments on written work were useful
32	8	7	6	9	6	4																			6.67	Instructor applies subject to the real world
28	6	8	5	6	6	7																			6.33	Workload is heavy
29	7	6	7	7	5	5																			6.17	Teacher is well prepared for each class
1	5	5	5	8	5	8																			6.00	Tests related to what is covered in class
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25	5	8	4	4	8	5																			5.67	Enjoyable and funny
26	3	7	3	7	5	8																			5.50	Instructor speaks in monotone voice
4	5	5	4	5	8	5																			5.33	Professor knows subject matter
10	6	4	5	8	5	4																			5.33	Clear, well structured presentations
8	5	3	3	7	7	6																			5.17	Instructor explains concepts well
16	3	5	6	2	6	9																			5.17	I learned a lot
27	5	6	5	4	4	6																			5.00	Thought provoking
15	2	4	7	5	4	8																			5.00	Tests are easy
30	3	7	6	5	3	6																			5.00	Class participation is encouraged
18	8	3	6	5	4	3																			4.83	Teaches from the book
17	6	3	8	3	3	6																			4.83	Tests/assignments are returned in a reasonable time
14	5	6	6	3	4	5																			4.83	Tests are mostly essay and short answer
7	4	9	3	4	5	3																			4.67	Professor is always available to help
19	7	2	7	7	2	3																			4.67	Lectures too much
24	8	6	5	3	3	2																			4.50	Challenging and demanding
9	4	6	3	8	2	3																			4.33	Teacher sets high academic standards
2	1	4	5	4	6	6																			4.33	Must do homework and be prepared for class
11	4	3	7	6	4	1																			4.17	Class is slow, tedious, boring
22	9	5	4	2	2	2																			4.00	Too much time spent on simple issues
12	4	5	2	4	1	7																			3.83	Instructor is punctual and attends class regularly
13	2	1	2	5	5	7																			3.67	Must take good note to be successful
3	3	5	4	3	3	4																			3.67	Talks over your head
6	6	2	1	1	8	2																			3.33	Instructor expectations are clear and consistent
23	2	2	2	2	6	4																			3.00	Profesor treats students with respect

Benefits/Concluding Remarks

- The coaching staff has easier access to the database, and can update the information more efficiently.
- The coaching staff can update the database from any off campus location via the World Wide Web.
- The coach can personally update the database without the Sports Information Director or without accessing the database itself.

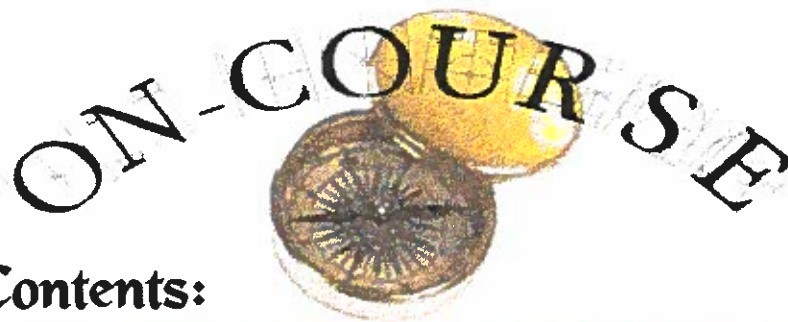


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- Partitioned Data Flow Diagram

Program Screens and Explanations

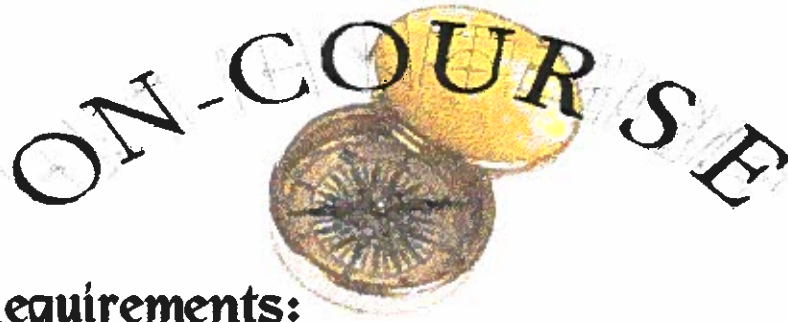
- Main Form
- Select Course Information
- Evaluation Form
- Change Settings

Database Information

- Tables
- Queries
- Reports
- Macros

Sample Reports

- Actual vs Desired
- Course Evaluation Summary – By Item
- Course Evaluation Summary – By Mean



System Requirements:

On-Course Evaluation Analysis System requires the use of Microsoft Access 2000 database software and Microsoft Excel for advanced reports, therefore the system requirements are only those needed to install and run these programs.

It is recommended that the user have the following system requirements:

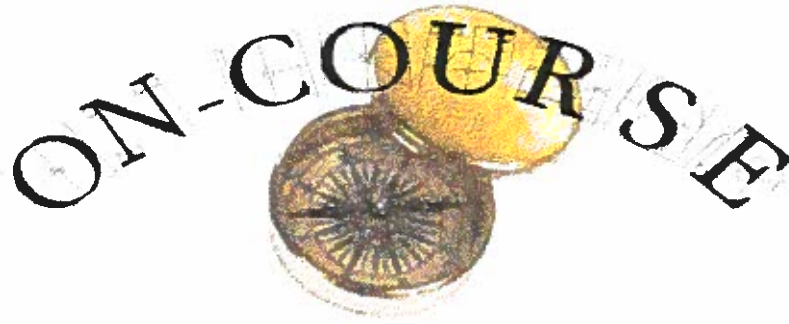
- Windows 95 or Higher
- 16 Megabytes of RAM or Higher and a Pentium Processor
- At least 12 Megabytes of Free Space on Hard Drive

Installation Instructions:

Floppy Disk Installation:

1. Create Directory C:\OnCourse
2. Go to the Windows Start Button
3. Select Run from the Start Menu
4. Type in a:
5. Open the self-extracting .zip file OnCourse.exe to C:\OnCourse
6. Copy the file MSVBM60.dll into the C:\Windows\System directory
7. Double Click on OnCourse.exe to launch program

A CD-ROM installation disk is available upon request



Project of:

Karen Thurston
106 Lantana Court
Simpsonville, SC 29681
(864) 675-0852 • (843) 870-6586

At the Request of:

Dr. Bill Bowers
Assoc. Professor of Management
Charleston Southern University
(843) 863-7908

Under the Supervision of:

Mr. Jim Roberts
Professor of Computer Science
Charleston Southern University
(843) 863-8083

Special Thanks to:

Mr. Robert Thurston
Mr. Todd Lea



Purpose:

On-Course uses Q-Sort Methodology (McKeown & Thomas, 1988) to evaluate University courses by automating the analysis of the evaluations, store and manipulate the evaluation data, and generate valuable reports on progress.

Background:

Charleston Southern University, along with many other institutions, has a system to evaluate the courses that are offered as well as the instructors that are teaching them. These evaluations serve to monitor the progress of the subject being evaluated to ensure optimum results for the students attending the university.

Currently at Charleston Southern University, the system of evaluations provides minimal information about the course, and does not coincide with a standard that is desired for each course and instructor. On-Course implements a system that can get data from student evaluations, fit it into a bell curve for easy analysis, store the data in an extensive database, and then generate reports based on how each course and instructor is performing as compared to the ideal standard.

**REQUEST FOR
INFORMATION
SYSTEM SERVICES**

DATE OF REQUEST	SERVICE REQUESTED FOR DEPARTMENT(S)
October, 14 2000	Business Department
SUBMITTED BY (key user contact)	EXECUTIVE SPONSOR (funding authority)
Name Dr. Bill Bowers	Name
Title Business Professor	Title
Office Jones Hall	Office
Phone (843) 863-7908	Phone

TYPE OF SERVICE REQUESTED:

- | | |
|--|---|
| ☐ Information Strategy Planning | ☐ Existing Application Enhancement |
| ☒ Business Process Analysis and Redesign | ☐ Existing Application Maintenance (problem fix) |
| ☒ New Application Development | ☐ Not Sure |
| ☐ Other (please specify _____) | |

BRIEF STATEMENT OF PROBLEM, OPPORTUNITY, OR DIRECTIVE (attach additional documentation as necessary)

Charleston Southern University currently has a system of course evaluations that allows students to evaluate the course that they are taking as well as the professor who is instructing the course. The information gained through these evaluations is very valuable to the professor, the department, and to the overall academic success of the university. This information can be used to see which courses and professors are succeeding in their work and which could use improvement. The problem with the current system is that there are no ideal standards that can be used to compare each class to the ideal classroom situation as seen by the university and the members of each department. Dr. William Bowers has requested a system be created that will allow students to evaluate each course and professor and then compare the results to an ideal scenario that has been previously established. With the new system, the data will be collected in such a way that it provides various statistical analysis reports that can give the university and the departments much more information than they have yet been able to receive from their current system of course evaluations.

BRIEF STATEMENT OF EXPECTED SOLUTION

The new system of course evaluations will use the Q Methodology (McKeown & Thomas, 1988) to collect and analyze students' responses. A set number of statements will be given and the student distributes the statements throughout several numbered columns whether or not the statement is most like or least like the class or professor that they are evaluating. The data will form a bell curve that can then be inputted into the computer system that will store the data in a database and create useful reports of information for the university and the department to evaluate. From these reports it will be clear to see how each class or professor compares to the ideal classroom setting that has been established by Charleston Southern University.

ACTION (ISS Office Use Only)☒ Feasibility assessment approvedAssigned to Karen Thurston☐ Feasibility assessment waived

Approved Budget \$ _____

Start Date 10/14/00 Deadline 5/9/01☐ Request delayed

Backlogged until date: _____

☐ Request rejected

Reason: _____

Authorized Signatures:

Chair, ISS Executive Steering Body_____
Project Executive Sponsor



PROBLEM STATEMENTS

PROJECT: Student Course Evaluation Analysis	PROJECT MANAGER: Karen Thurston
CREATED BY: Karen Thurston	LAST UPDATED BY: Karen Thurston
DATE CREATED: 10/14/00	DATE LAST UPDATED: 10/14/00

Brief Statements of Problem, Opportunity, or Directive	Urgency	Visibility	Annual Benefits	Priority or Rank	Proposed Solution
1. Current system does not compare courses or professors to a set standard.	ASAP	High	Will set a standard for all classes.	1	Use Q Methodology that will have an ideal evaluation that is desired of the course and compare students' responses to the ideal.
2. Data needs to be processed using statistical analysis to provide useful information to the university and the departments.	4 months	High	Will provide detailed analysis of the data.	2	Use the data, which forms a bell curve, and analyze it using several statistical equations.
3. Need an automated system to analyze course evaluations.	7 months	High	Will speed process and give accurate results with less effort.	3	Create automated system that will process the data and generate reports.

FIGURE 4.8



Project Feasibility Assessment Report

I. Executive Summary

A. Summary of Recommendation

On September 15, 2000, Dr. Bill Bowers contacted Greenstone Consulting in regards to a project that would greatly enhance the current system of student course evaluations at Charleston Southern University. Greenstone Consulting will, using a new method of evaluation, create an automated system that will store evaluation data into a database and create reports of the data gathered from the evaluations. The new system will use Q Methodology to compare the students' views of the course and the professor to that of the university's ideal of what the course and professor should be. The new system will take this data, allow it to be inputted for each student, automatically store it in a database, and produce reports that will contain valuable information to the university and each department.

B. Brief Statement of Anticipated Benefits

This new system, once implemented, will provide immediate benefits not only to the university, but also to each individual department, each professor, and possibly even to the students. The data will be analyzed to compare how different the course is to the ideal set by the university. It will also show how different courses compare, and which methods and styles of teaching prove most effective to meeting the university's standard. The major benefit will be that problems and successes can be easily spotted and trouble areas can be fixed for better overall academic success.

C. Brief Explanation of Report Contents

The contents of this report following this summary include a detailed description of the Student Course Evaluation project. It will include Greenstone Consulting, along with Dr. Bill Bowers, recommendations and ideas to solve the problems involved in the project. This report will also contain the project plan and schedule of each step of the project.

II. Background Information

A. Brief Description of Project Request

Dr. Bill Bowers of Charleston Southern University has requested of Greenstone Consulting that an automated computer system be created that can take the students evaluations and produce useful reports for the university and the individual departments. The current system is not set up to compare each course and professor to a standard set by the university, and therefore does not give as accurate of information about where the weaknesses are and which areas need most improvement. The statistics involved with the new system would give the most accurate information possible that could be used for improvements of the courses as well as the professors. The goal is to have the system complete by May of 2001. The goal of this system is to make the courses and operations at Charleston Southern University more effective by providing an accurate evaluation of each course and professor by the students of the university.

B. Brief explanation of the survey phase activities

Greenstone Consulting met with Dr. Bill Bowers of Charleston Southern University in September 2000. Dr. Bowers will be the main contact for the project, and Karen Thurston will be the project manager for Greenstone Consulting. Karen will work with the assistance of Mr. Jim Roberts to design and implement this system as requested by Dr. Bill Bowers. The request was made by Dr. Bowers and evaluated by Karen Thurston. After evaluating the problems, opportunities, and objectives of this request, it has been decided by Greenstone Consulting to go forth with this request. An allotted time of eight months will be given to the completion of this project, with the final deadline being May, 2001.

III. Detailed Recommendation

A. Narrative Recommendation

1. Immediate fixes

Due to the fact that Student Course Evaluations are only performed once a semester, the university will continue with their current system of evaluating courses until the new system is completed and adopted by the university. Until that time there will be no immediate fixes to the system.

2. Enhancements

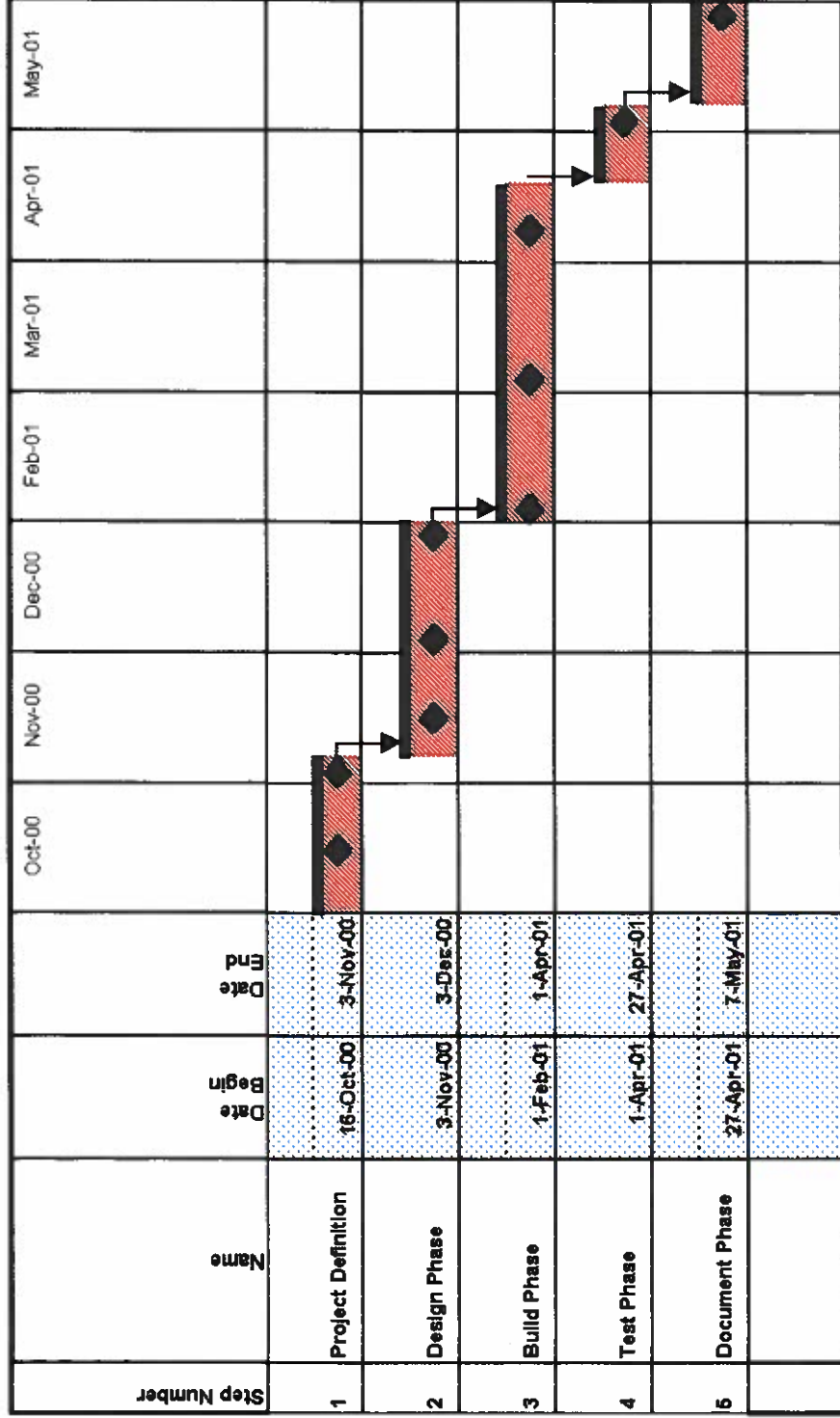
The enhancements made to the old system will be that of using Q Methodology to evaluate the courses instead of only a few questions answered using 1-5 to rate most like or least like the course. The Q Methodology will use 32 statements that will be divided amongst 9 categories, forcing the answers into a bell curve.

3. New Systems Development

Currently there is no automated way of reporting the results of the student evaluations. The new system will provide several options of analyzing the data, including sorting data by the mean, calculating the correlation between different classes, and various other ways. The new system will produce reports from the input that will reflect accurately the overall perception of the course and professor by the students that are enrolled in that course.

B. Project Plan

- 1. Initial Master Project Plan (Phase Level - Gantt Chart)**
- 2. Detailed Plan for the Study or Definition Phase (Gantt Chart)**



Step Number	Project Definition and Design Phase	Begin Date	End Date	Oct-00	Nov-00	Dec-00
	Request for Systems Services					
1	Problem Statement Matrix	16-Oct-00	23-Oct-00			
2	Feasibility Assessment Report	16-Oct-00	23-Oct-00			
3	Gantt Charts Prob, Opp, Obj & Constr Matrix	23-Oct-00	6-Nov-00			
4	E-R Diagram Entity/Definition Matrix	6-Nov-00	27-Nov-00			
6	Data Flow Diagrams E-R List & Decision Tables	28-Nov-00	1-Dec-00			
6	Locatin Connectivity Diagram Candidate & Feasibility Matrix	28-Nov-00	1-Dec-00			
7	Systems Proposal Report	1-Dec-00	10-Dec-00			



PROBLEMS, OPPORTUNITIES, OBJECTIVES AND CONSTRAINTS MATRIX

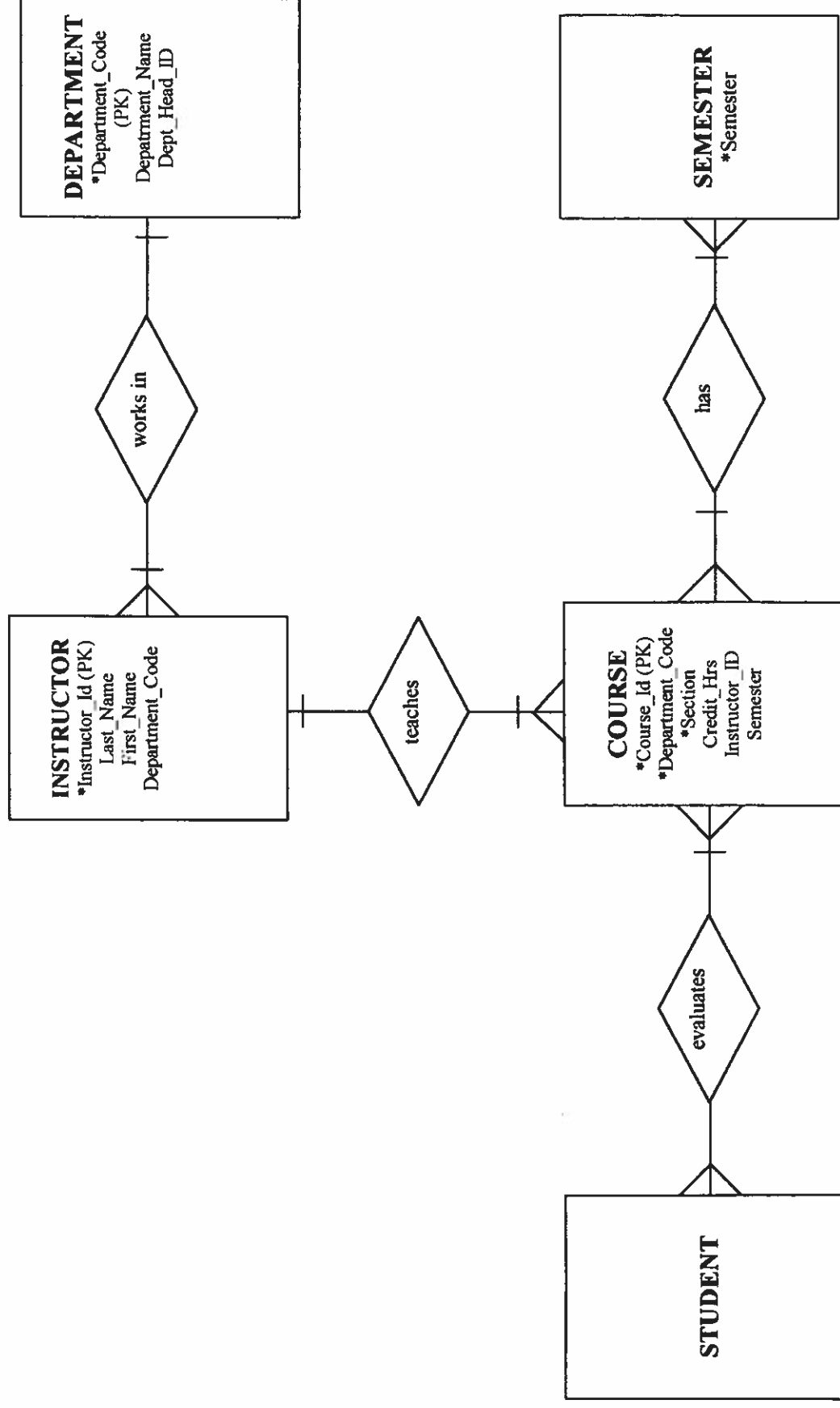
Project:	Student Course Evaluation Analysis	Project Manager:	Karen Thurston
Created by:	Karen Thurston	Last Updated by:	Karen Thurston
Date Created:	10/22/00	Date Last Updated:	10/22/00

CAUSE AND EFFECT ANALYSIS		SYSTEM IMPROVEMENT OBJECTIVES	
Problem or Opportunity	Causes and Effects	System Objective	System Constraint
1. There is no automated program to generate reports from data collected by student course evaluations. 2. The current system of evaluations is not very thorough and does not give enough information to derive detailed reports.	1. The current system of evaluations uses a basic form with only a few questions. 2. The data received by these evaluations is not in a format that can be used for simple statistical calculations. 3. Reports need to be generated that give the university and the departments solid data on each course and each professor, as seen by the students.	1. To use a new system of course evaluations that will implement Q Methodology. 2. To retrieve the data from these evaluations and perform statistical calculations on this data. 3. To create a system to automate the process of analyzing the results. 4. To produce reports that will reflect the students' views on each course and professor compared to the benchmark established by the university.	1. The automated system must eventually be able to be multi-user 2. The system should be user friendly and provide an input screen and several report choices based on the input. 3. Desired completion date of May 2001.

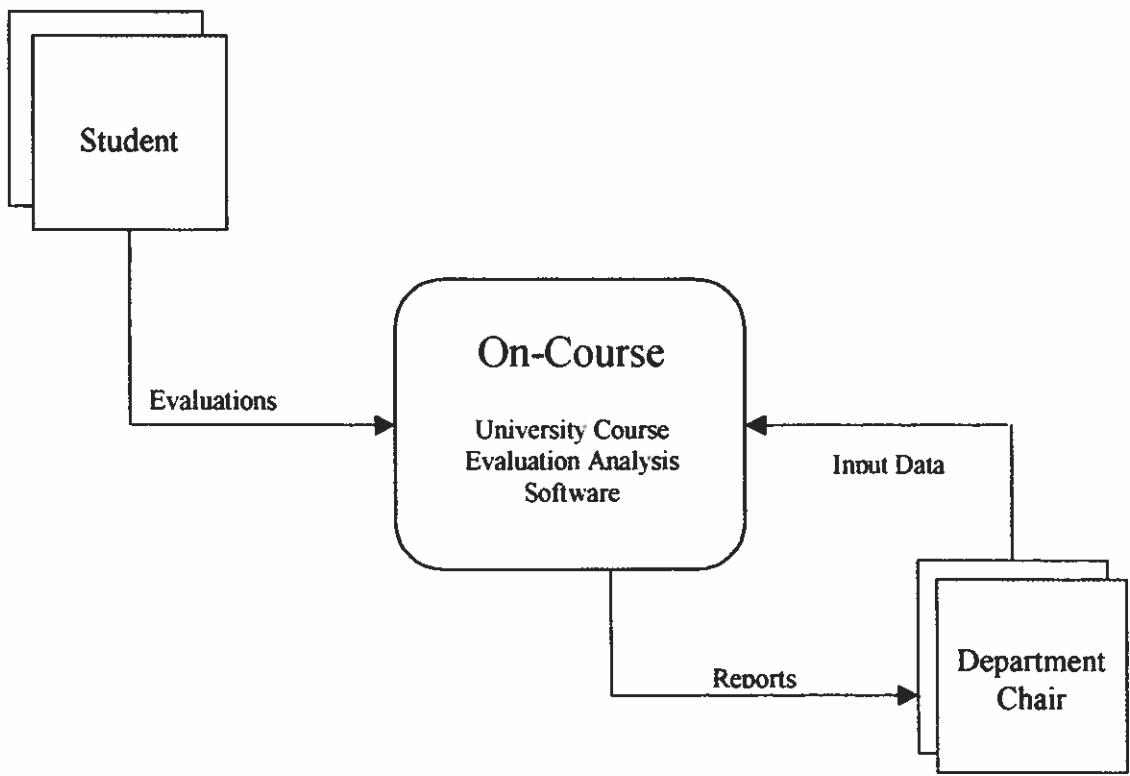
Entity/Definition Matrix

ENTITY	BUSINESS DEFINITION
STUDENT	A person enrolled in a course at the university.
COURSE	A course taught by the professor that has been approved by the university.
PROFESSOR	An employee of the university who works in a department and teaches a course.
SEMESTER	A period of time during which a course is offered by the university.
DEPARTMENT	A department is established by the university and is comprised of professors who teach courses of a particular subject matter.

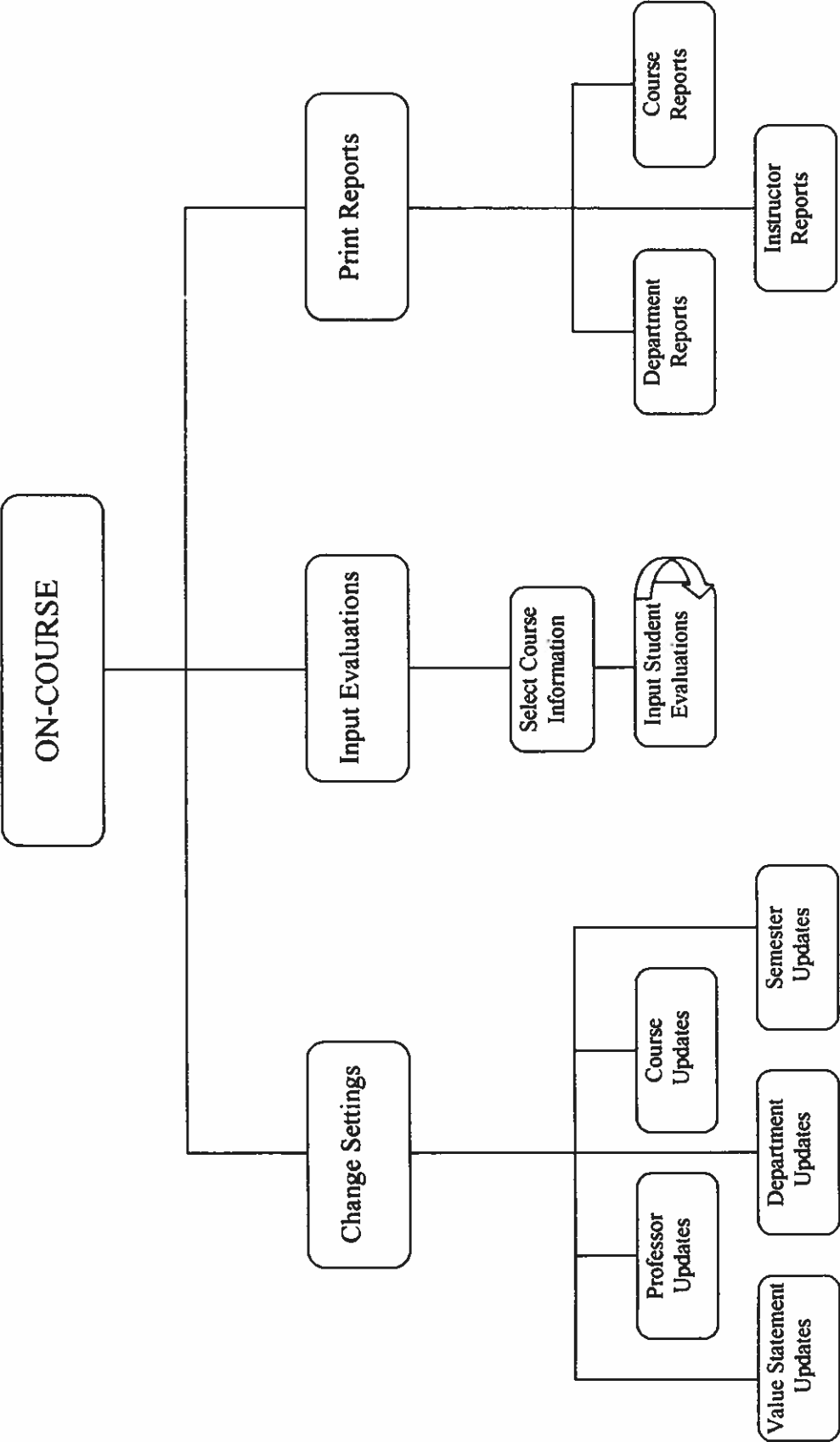
Entity Relationship Diagram



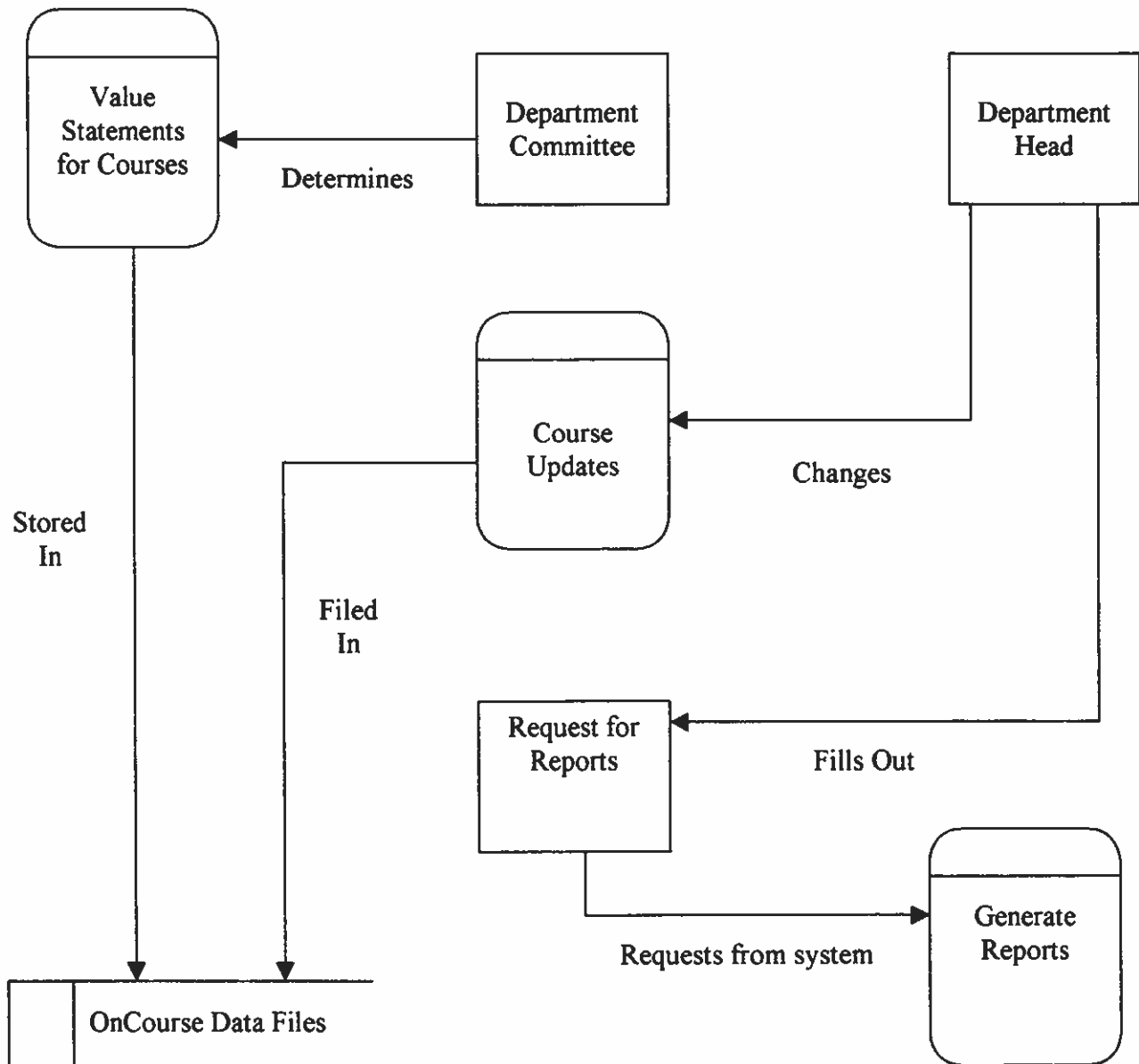
CONTEXT DIAGRAM



DECOMPOSITION DIAGRAM



SYSTEM DIAGRAM



EVENT LIST

Event Description	Trigger (Inputs)	Responses (Outputs)
Student Evaluates Course	Data-Entry program stores their data	Updates the database with the student's evaluation
Department Requests Analysis	Request made of the system to analyze the data for a course	Reports are made available for printing or viewing
Course Report Generated	Analysis is requested for all evaluations for a particular course	Statistics for the evaluations of that course are given
Courses Change	A course is added to the curriculum	The course database is updated
Departments Change	A department is added or information changes	The department database is updated
Professors Change	A professor is added or changed from a course	The professor database is updated
Value Statements are Changed	An authorized user (a department head) creates a new set of value statements	The value statements stored for that department are updated

CANDIDATE MATRIX

Characteristics	Candidate 1	Candidate 2	Candidate 3
Portion of System Computerized Brief description of that portion of the system that would be computerized in this candidate.	COTS software package designed to do basic statistical analysis of data.	After the data is manually inputted, the system would analyze the data and produce customized reports.	The system would have a computerized method of evaluating the courses and then would analyze the data and produce requested reports.
Benefits Brief description of the business benefits that would be realized for this candidate.	Would be a simplified and cost effective way to get the results.	Would be a customized program to give quick and informative reports that would be useful for the departments and the university.	Would eliminate the paperwork necessary for course evaluations and produce quick and informative reports that would be useful for the departments and the university.
Servers and Workstations A description of the servers and workstations needed to support this candidate.	Pentium Workstations that could load and run the software.	Pentium Workstations with Windows 95 or higher, Microsoft Access and Excel, and a network that could run the software.	Same as Candidate 2
Software Tools Needed Software tools needed to design and build the candidate (e. g., database management system, emulators, operating systems, languages, etc.). Not generally applicable if applications software packages are to be purchased.	No Tools Needed, Software Package Purchased	Microsoft Access and Microsoft Excel with a Visual Basic Front End.	Microsoft Access, Microsoft Excel, Microsoft Visual Basic and Java Programming.
Application Software A description of the software to be purchased, built, accessed, or some combination of these techniques.	Package Solution	Custom Solution	Custom Solution
Method of Data Processing Generally some combination of: on-line, batch, deferred batch, remote batch, and real-time.	Batch	Same as Candidate 1	Same as Candidate 1
Output Devices and Implications A description of output devices that would be used, special output requirements, (e.g. network, preprinted forms, etc.), and output considerations (e.g., timing constraints).	Preprinted Forms developed by the software that report the results of the statistical equations. Printed on HP5SI LAN Laser Printers.	Customized forms that would print desired reports on HP5SI LAN Laser Printers.	Same as Candidate 2
Input Devices and Implications A description of Input methods to be used, input devices (e.g., keyboard, mouse, etc.), special input requirements, (e.g. new or revised forms from which data would be input), and input considerations (e.g., timing of actual inputs).	Keyboard, Mouse, Data Forms (Evaluations)	Same as Candidate 1	Keyboard, Mouse
Storage Devices and Implications Brief description of what data would be stored, what data would be accessed from existing stores, what storage media would be used, how much storage capacity would be needed, and how data would be organized.	Evaluation data would be stored on the PC hard drive that it was input to.	All evaluations, course, and department information would be stored in the main database files. The data would be kept on the server, but also distributed through the department workstations on backup systems for safety and security precautions.	Same as Candidate 2

FEASIBILITY MATRIX

Feasibility Criteria	Wt.	Candidate 1	Candidate 2	Candidate 3
Operational Feasibility Functionality. A description of to what degree the candidate would benefit the organization and how well the system would work. Political. A description of how well received this solution would be from both user management, user, and organization perspective.	60%	Would be a pre-purchased software package and could not be customized for their needs. Score: 60	Would support the evaluations, department information, and course information. Would analyze the data and produce custom reports. Score: 90	Same as Candidate 2 with an computerized evaluation process added. Score: 100
Technical Feasibility Technology. An assessment of the maturity, availability (or ability to acquire), and desirability of the computer technology needed to support this candidate. Expertise. An assessment to the technical expertise needed to develop, operate, and maintain the candidate system.	20%	Would be subject to upgrades from the pre-purchased software company. This Candidate would require a staff with the knowledge to install and maintain these upgrades. Score: 60	User friendly and designed for university faculty in mind. Would need minimal maintenance and could be upgraded upon request by Greenstone Consultants. Score: 100	Same as Candidate 2 only additional technical skill would be needed to maintain and upgrade the computerized evaluation system along with training and monitoring the usage. Score: 80
Economic Feasibility Cost to develop: Payback period (discounted): Net present value: Detailed calculations:	0%	Approximately \$ Approximately \$ Approximately \$ Score: unknown	Approximately \$ Approximately \$ Approximately \$ Score: unknown	Approximately \$ Approximately \$ Approximately \$ Score: unknown
Schedule Feasibility An assessment of how long the solution will take to design and implement.	20%	Less than 3 months Score: 100	Less than 8 months Score: 100	More than 1 year Score: 60
Ranking:	100%	68	94	88



SYSTEM PROPOSAL

I. Introduction

A. Purpose of the report

The purpose of this report is to review and select a candidate solution from those that have been proposed for the system that has been requested. This report will give information and explanation for the proposed candidates and will recommend the best choice, as chosen by Greenstone Consultants.

B. Background of the project leading to this report

An analysis of the systems request has been done, and Greenstone has determined that there are several approaches to fulfilling the request. Three of the candidates have been identified in the candidate matrix. The information is based on preliminary work done by Greenstone to evaluate the system and the needs of this project.

C. Scope of the project

This project involves a system that will automatically evaluate the results of student course evaluations. The scope of the project could be expanded to include an automated system of evaluations or could be limited to simply an analysis for the current system of evaluations.

D. Structure of the report

This report will give the recommended solution to the system request based on the work done by Greenstone Consultants and the operational, technical, economic, and schedule feasibility of the candidate based on the needs of the university.

II. Tools and techniques used

A. Solution Generated

Having reviewed the three candidates and their feasibility, Greenstone is recommending candidate 2 as the best solution to the operational, technical, economic, and schedule needs of the university. The system needs to be customized to meet the needs of the university and to produce customized reports upon request from the departments or the university. The system also needs to be easily maintained and used by the faculty of the university. Although some customized systems would meet these requirements, it is also important that the system be economically feasible and done in a reasonable amount of time.

B. Feasibility analysis (cost-benefit)

Candidate 2 will provide all the benefits of a system that is customized for the university's needs. It is the ideal candidate for a system that will analyze the student's course evaluations and produce reports that will benefit the professors, the departments, and the university. Candidate 2 will be created and supported by Greenstone Consulting. This system can be maintained and operated with very little technical knowledge necessary to the users. This candidate also complies with the economic feasibility of the university and can be completed and implemented within a reasonable amount of time.

III. Information systems requirements

This candidate would produce customized software that could be installed on any Pentium workstation that could support Microsoft Access and Excel or on a server and could be accessed from any of these workstations. The workstations could store

the software as well as the databases needed, and would require connection to a LAN printer to produce the customized reports.

IV. Alternative solutions and feasibility analysis

The alternative solutions are found in candidates 1 and 3. These candidates propose systems that are either below the operational desires of the university or above the economic and schedule feasibility. Candidate 1 proposes a pre-packaged statistical system that could be purchased and used to run basic statistical calculations. This system would not produce the customized results that are desired, and would not be flexible to changes and upgrades from Greenstone. This system would be cost effective and timely, but the benefits would be so few that the cost would not outweigh it.

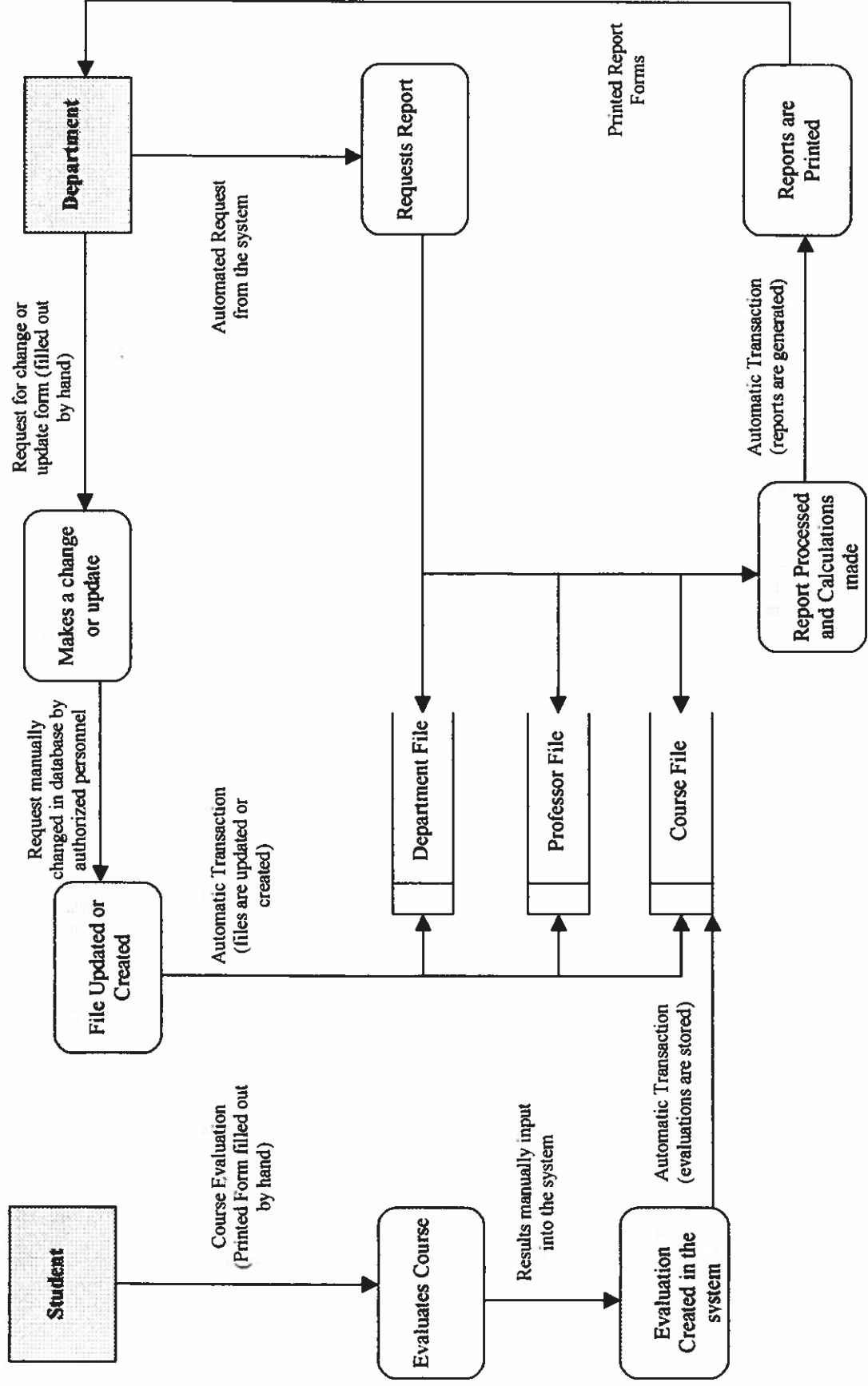
Candidate 3 would be an extensive system that would meet all the needs requested system. It would require advanced programming which would make it an expensive and time consuming project that somewhat outweighs the benefits that would come from this advanced system. It would also require more technical training and maintenance because there would be a wide scope of users for the computerized evaluation system.

V. Recommendations

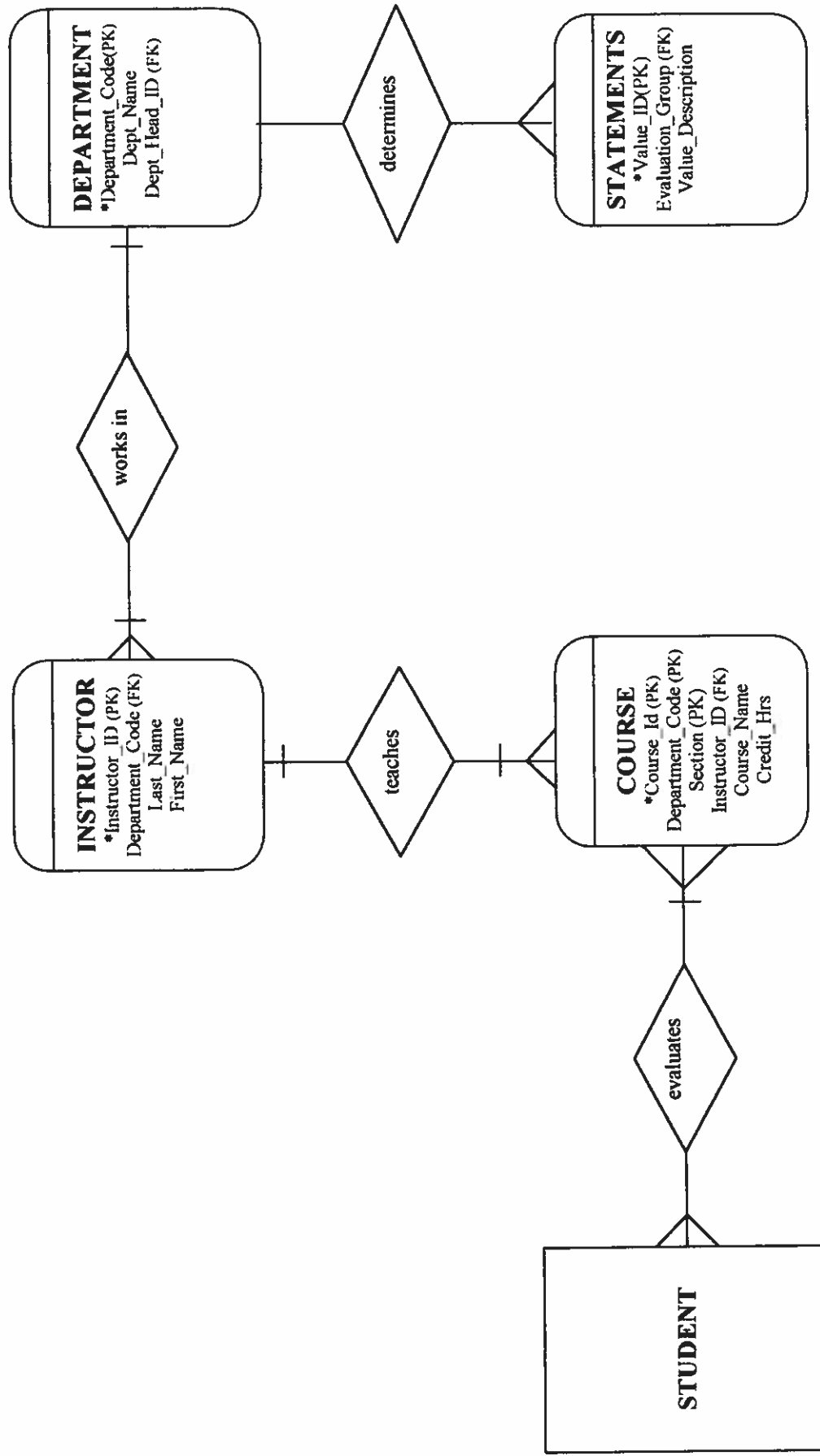
Based on the analysis of the three candidates, Greenstone Consulting is recommending candidate 2 as the best system proposal to meet the needs of the university. Candidate 2 is the best combination of operational, technical, economic and schedule feasibility amongst the three candidates. It is a customized system that can be easily used and maintained and will provide the results that are desired.

Candidate 2 is the top choice by Greenstone to meet the needs of the university.

PHYSICAL DATA FLOW DIAGRAM



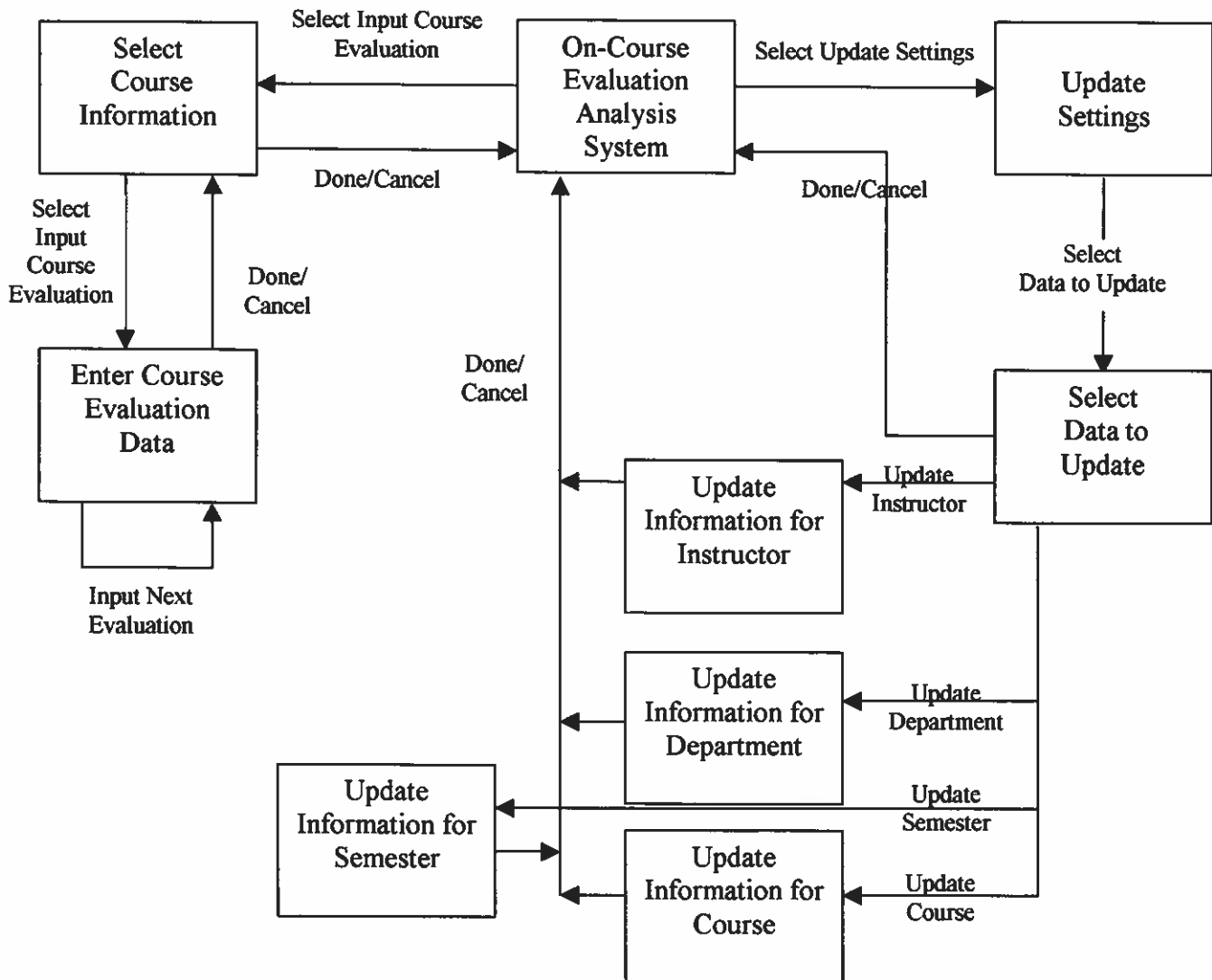
Normalized Relations Diagram



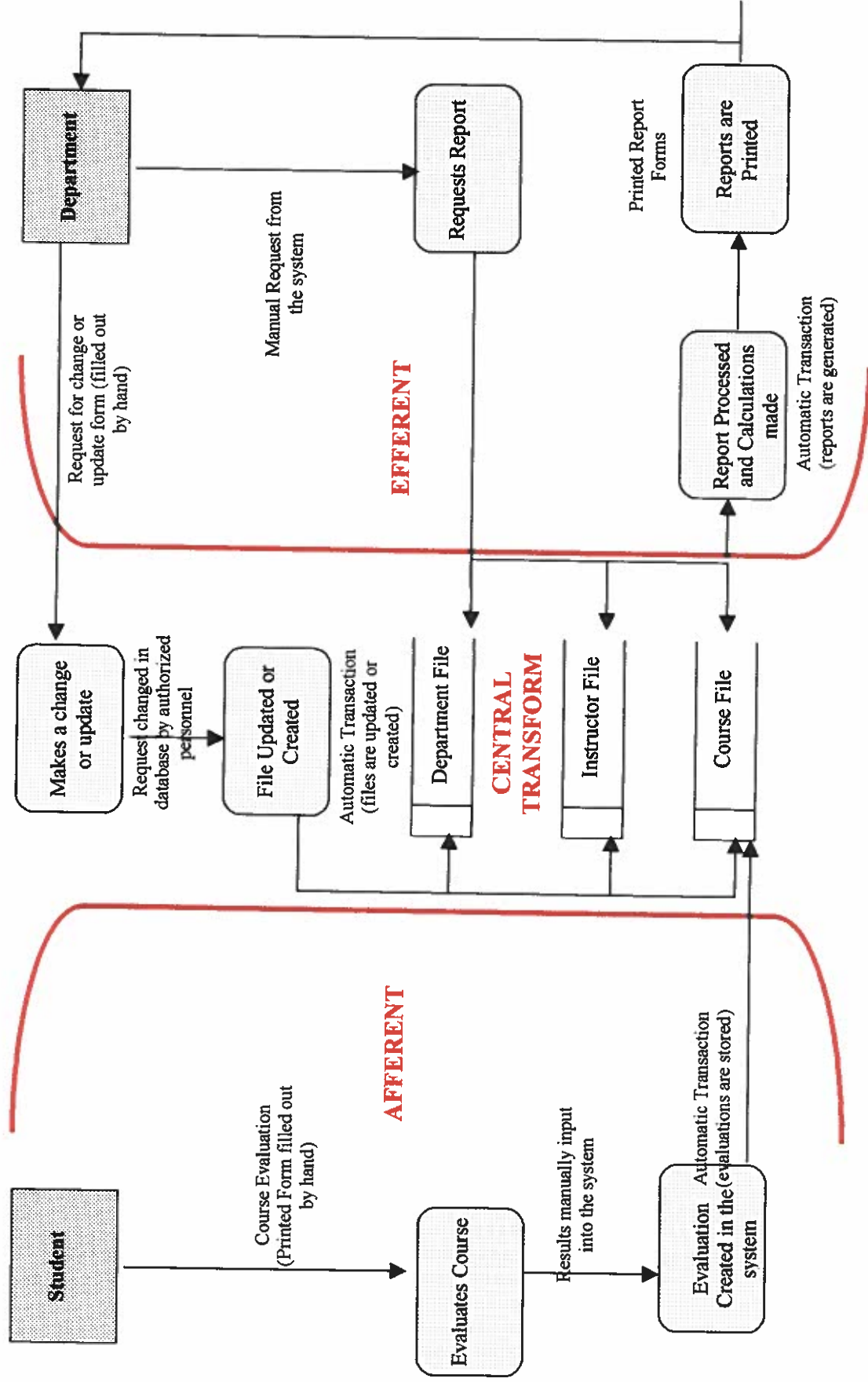
COMPLETE TABLE DEFINITIONS

Table Name	Field Name	Type	Size	Description	Mask
Course	Department_Code (PK)	VARCHAR	4		AAAA
	Course_Id (PK)	CHAR	3		###
	Section (PK)	CHAR	2		##
	Semester (FK)	VARCHAR	10		
	Instructor_Id (FK)	CHAR	3		###
	Course_Name	VARCHAR	30		
	Credit_Hrs	CHAR	1		#
Instructor	Instructor_ID (PK)	CHAR	3		###
	Department_Code (FK)	VARCHAR	4		AAAA
	Last_Name	VARCHAR	20		
	First_Name	VARCHAR	20		
Department	Department_Code (PK)	VARCHAR	4		AAAA
	Dept_Head_Id (FK)	CHAR	3		###
	Dept_Name	VARCHAR	20		
Value Statements	Value_Id (PK)	CHAR	2		##
	Evaluation_Group	VARCHAR	4		AAAA
	Value_Description	VARCHAR	50		

STATE TRANSITION DIAGRAM



PARTITIONED DATA FLOW DIAGRAM



Main Form

Code for frmMain.frm

```
Private Sub cmdExit_Click()
```

```
' Closes all open screens and terminates the program  
Unload Me
```

```
End Sub
```

```
Private Sub cmdInputEvaluation_Click()
```

```
'Opens frmInputEvaluation1 which allows the user to select the course information  
Load frmInputEvaluation1  
frmInputEvaluation1.Show  
frmInputEvaluation1.Refresh
```

```
End Sub
```

```
Private Sub cmdRequestReports_Click()
```

```
'Opens frmReports which allows the user to select report options  
Load frmReports  
frmReports.Show
```

```
End Sub
```

```
Private Sub cmdSettings_Click()
```

```
' Opens frmSetting which allows the user to add data to the OnCourse data tables  
Load frmSettings  
frmSettings.Show
```

```
End Sub
```

Select Course Information

Instructor:

Semester:

Department Code:

Course Number:

Section:

 Main Menu

 Input Evaluation

Select Course Information

frmInputEvaluation1.frm

This form is pulled up when the user selects "Input Evaluation" from the Main form. This screen consists of 4 combo boxes and a text box.

When the user selects an Instructor_ID from the Instructor combo box, the form automatically fills in the department code for that particular instructor. All subsequent combo boxes are locked until the previous selection is made. For instance, the user is unable to choose a section until a course number has been selected.

Each combo box on this screen is dependant on the previous boxes in order to narrow down the selection. When a semester is chosen, the course numbers listed are only those taught by the selected instructor during the selected semester. The section is again narrowed down to only those of the selected course number.

If the user wishes to change the data that has been selected, re-selecting an Instructor_ID from the Instructor box will update the department code and clear out all previous selections.

When all course information has been entered, the user may select "Input Evaluation" to begin entering the students' evaluations.

"Main Menu" will close this screen and return back to the Main form.

Select Course Information Form

Code for frmInputEvaluation1.frm

Public Sub Fill(typeFill As String)

'Sets up cases that will be called to fill each combo box at the appropriate time

Select Case typeFill

Dim cnn As New ADODB.Connection 'establishes ADODB connection to the database

Dim rs As New ADODB.Recordset

Dim cnstr As String

Dim sqlStr As String

Dim returnVal As Integer

Dim instrCode As String

SAMPLE CODE FOR CASE "INSTRUCTOR" – repeated and modified for all other cases (Department, Semester, Course, Section)

'FILL INSTRUCTOR COMBO BOX

Case "Instructor"

returnVal = 0

sqlStr = "Select * from Instructor"

cnstr = "Driver={Microsoft Access Driver (*.mdb)};Dbq=c:\OnCourse\oncourse.mdb;Uid=;Pwd="

With cnn

.Open cnstr 'opens connection to On-Course database

End With

With rs

.CursorType = adOpenForwardOnly

.CursorLocation = 3

.LockType = adLockOptimistic

.Open sqlStr, cnn

End With

'Adds the records from Instructor table to the combo box

If rs.RecordCount > 0 Then

Do Until rs.EOF

frmInputEvaluation1.cmbInstructor.AddItem (rs.Fields("Instructor_ID").Value)

rs.MoveNext

Loop

Else

yn = MsgBox("There was no data entered in the Instructor table.", vbCritical + vbOKOnly, "Critical Error")

End If

rs.Close

cnn.Close

End Select

End Sub

Private Sub cmbCourse_Click()

'Enables and fills Section combo box once course has been selected

frmInputEvaluation1.cmbSection.Enabled = True

frmInputEvaluation1.cmbSection.Clear

Call Fill("Section")

End Sub

Code for frmInputEvaluation1.frm (cont...)

```
Private Sub cmbInstructor_Click()  
'Enables and fills Semester and Department once Instructor_ID had been selected  
frmInputEvaluation1.cmbSemester.Enabled = True  
frmInputEvaluation1.cmbSemester.Clear  
frmInputEvaluation1.cmbCourse.Clear  
frmInputEvaluation1.cmbSection.Clear  
Call Fill("Semester")  
Call Fill("Dept")  
End Sub
```

```
Private Sub cmbSection_Click()  
'Enables the user to select "Input Evaluation" once all fields have been selected  
frmInputEvaluation1.cmdInputEvaluation.Enabled = True  
End Sub
```

```
Private Sub cmbSemester_Click()  
'Enables and fills Course combo box once the semester has been selected  
frmInputEvaluation1.cmbCourse.Enabled = True  
frmInputEvaluation1.cmbCourse.Clear  
frmInputEvaluation1.cmbSection.Clear  
Call Fill("Course")  
End Sub
```

```
Private Sub Form_Activate()  
'Enables and fills the Instructor combo box when the form is initialized  
Dim cnn As New ADODB.Connection  
Dim rs As New ADODB.Recordset  
Dim cnstr As String  
Dim sqlStr As String  
Dim returnVal As Integer  
Call Fill("Instructor")  
End Sub
```

```
Private Sub cmdInputEvaluation_Click()  
'Loads the frmEval that will allow the user to input the evaluations for the selected course  
Load frmEval  
frmEval.Show  
End Sub
```

```
Private Sub cmdReturn_Click()  
'Returns to the Main Menu  
Unload Me  
Load frmMain  
frmMain.Show  
End Sub
```

ON-COURSE

NEUTRAL or
UNIMPORTANT

LEAST like course

MOST like course

lstBox1 lstBox2 lstBox3 lstBox4 lstBox5 lstBox6 lstBox7 lstBox8 lstBox9

1

3

4

5

6

5

4

3

1

Student

Instructor

Course

Semester

lstValue

Input Evaluation

Main Menu

Input Next
Evaluation

Evaluation
Form

frmEval.frm

This form is brought up from the Select Course Information Screen when the user selects "Input Evaluation". The course information that the user has selected will automatically appear in the text boxes on the bottom left of the screen. The user MUST enter a student number (beginning with 1) for each student in a particular course before the data from this screen can be saved into the On-Course database.

The values contained in the list box at the bottom center of the screen correspond to the value statements 1 – 32 that were given to evaluate the course. The students have placed each value in one of the list boxes 1-9. List box 1 being LEAST like the course and list box 9 being MOST like the course. The values from the bottom list box must be dragged and dropped into the appropriate list box corresponding with the students' evaluations.

When a value is moved from the bottom box into one of the top list boxes, the value will be removed as a selection from the list. This is to ensure that all values are used once and only once. When all the values have been dragged into the appropriate list box and the student number has been entered, the user must click "Input Next Evaluation" in order to save the data into the On-Course database and bring up a clear screen for the next evaluation.

Input Evaluation Form

Code for frmEval.frm

```

Private Sub cmdInput_Click()
'Inputs the student evaluations into the data table of the On-Course database
Dim cnn As New ADODB.Connection
Dim rs As New ADODB.Recordset
Dim cnstr As String
Dim sqlStr As String
Dim returnVal As Integer
'FILL DATA IN DB
    cnstr = "Driver={Microsoft Access Driver (*.mdb)};Dbq=c:\OnCourse\oncourse.mdb;Uid=;Pwd="
    With cnn
        .Open cnstr      'establishes a connection to On-Course database
    End With
    returnVal = 0
    sqlStr = "Select * from data where student='XXXXXXXXXX'"
    With rs
        .CursorType = adOpenForwardOnly
        .CursorLocation = adUseClient
        .LockType = adLockOptimistic
        .Open sqlStr, cnn
    End With
'Verifies that the user has entered the student number
    If Trim(frmEval.txtStudent.Text) = "" Then
        yn = MsgBox("You must enter a Student number.", vbOKOnly, "Data Error")
        Exit Sub
    End If
    If rs.RecordCount = 0 Then
        'This loops through listbox 1 and stores the info from the box
        ***SAMPLE CODE FOR STORING DATA FROM THE LISTBOX – repeated and modified for
        lstBox 3 through 9***
        For I = 0 To frmEval.lstBox1.ListCount - 1
            rs.AddNew
            rs.Fields("code").Value = Trim(frmEval.lstBox1.List(I))
            rs.Fields("box_number").Value = "1"
            rs.Fields("row_number").Value = I
            rs.Fields("course").Value = Trim(frmEval.txtCourse.Text)
            rs.Fields("student").Value = Trim(frmEval.txtStudent.Text)
        Next I
        For I = 0 To frmEval.lstBox2.ListCount - 1
            rs.AddNew
            rs.Fields("code").Value = Trim(frmEval.lstBox2.List(I))
            rs.Fields("box_number").Value = "2"
            rs.Fields("row_number").Value = I
            rs.Fields("course").Value = Trim(frmEval.txtCourse.Text)
            rs.Fields("student").Value = Trim(frmEval.txtStudent.Text)
        Next I
        rs.Update
    End If
    rs.Close

```

Code for frmEval.frm (cont...)

```
cnn.Close
Unload Me
Load frmEval
frmEval.Show
End Sub
```

```
Private Sub Form_Activate()
Dim cnn As New ADODB.Connection
Dim rs As New ADODB.Recordset
Dim cnstr As String
Dim sqlStr As String
Dim returnVal As Integer
'Keeps the course information stored until the "Select Course Information" screen is changed
txtInstructor.Text = Trim(frmInputEvaluation1.cmbInstructor.Text)
txtCourse.Text = Trim(frmInputEvaluation1.txtDepartment.Text) + " " +
Trim(frmInputEvaluation1.cmbCourse.Text) + " " + Trim(frmInputEvaluation1.cmbSection.Text)
txtSemester.Text = Trim(frmInputEvaluation1.cmbSemester.Text)
returnVal = 0
sqlStr = "Select * from Statements"
cnstr = "Driver={Microsoft Access Driver (*.mdb)};Dbq=c:\OnCourse\oncourse.mdb;Uid=;Pwd="
With cnn
    .Open cnstr
End With
With rs
    .CursorType = 2
    .CursorLocation = 3
    .LockType = 3
    .Open sqlStr, cnn
End With
'Fills the lstValue list box
If rs.RecordCount > 0 Then
    Do Until rs.EOF
        frmEval.lstValueSt.AddItem (rs.Fields("Value_Id").Value)
        rs.MoveNext
    Loop
Else
    yn = MsgBox("There was no data entered in the Statements table.", vbCritical + vbOKOnly, "Critical
Error")
End If
rs.Close
cnn.Close
End Sub
```

SAMPLE CODE FOR DRAG & DROP – modified and repeated for all 9 list boxes

```
'Allows the user to drag & drop the values into the appropriate list boxes
Private Sub lstBox1_DragOver(Source As Control, X As Single, Y As Single, State As Integer)
Dim curVal As String
Dim I As Integer
Dim curValIndex As Integer
curValIndex = Source.ListIndex
curVal = Trim(Source.List(Source.ListIndex))
```

Code for frmEval.frm (cont...)

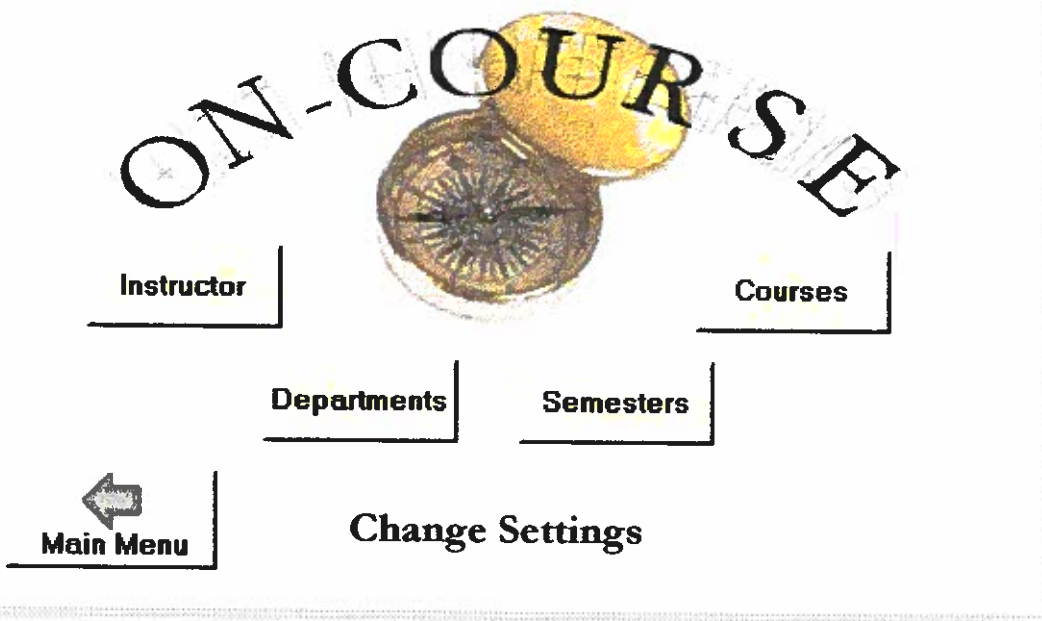
```
If Trim(curVal) = "" Then
    Exit Sub
End If
For I = 0 To frmEval.lstBox1.ListCount - 1 ' Loop through list.
    If Trim(curVal) = frmEval.lstBox1.List(I) Then
        Exit Sub
    End If
Next I
If frmEval.lstBox1.ListCount < 1 Then
    If State = vbEnter Then
        frmEval.lstBox1.AddItem (curVal)
        Source.RemoveItem (curValIndex)
    End If
End If
Source.Drag (0)
End Sub
```

'Picks up the value that the mouse is clicked on

```
Private Sub lstBox1_MouseDown(Button As Integer, Shift As Integer, X As Single, Y As Single)
If frmEval.lstBox1.List(frmEval.lstBox1.ListIndex) = "" Then
    frmEval.lstBox1.Drag (0)
Else
    frmEval.lstBox1.DragMode = 0
    frmEval.lstBox1.Drag (1)
End If
End Sub
```

```
Private Sub lstBox1_MouseUp(Button As Integer, Shift As Integer, X As Single, Y As Single)
    frmEval.lstBox1.DragMode = 0
    frmEval.lstBox1.Drag (2)
End Sub
```

```
Private Sub lstValueSt_DragOver(Source As Control, X As Single, Y As Single, State As Integer)
    Dim curVal As String
    Dim I As Integer
    Dim curValIndex As Integer
    curValIndex = Source.ListIndex
    curVal = Trim(Source.List(Source.ListIndex))
    For I = 0 To frmEval.lstValueSt.ListCount - 1 ' Loop through list.
        If Trim(curVal) = frmEval.lstValueSt.List(I) Then
            Exit Sub
        End If
    Next I
    'removes the value from the lstValue once is has been put into another list box
    If frmEval.lstValueSt.ListCount < 1 Then
        If State = vbEnter Then
            frmEval.lstValueSt.AddItem (curVal)
            Source.RemoveItem (curValIndex)
        End If
    End If
    Source.Drag (0)
End Sub
```



Change Settings

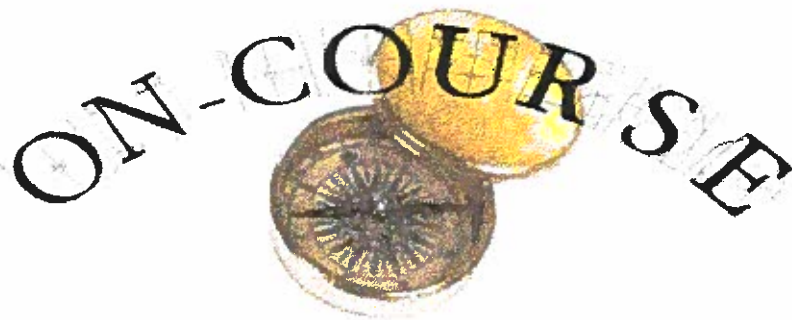
frmSettings.frm

When the user selects “Change Settings” from the Main form, this screen appears. The feature to change settings allows the user to automatically add data to the On-Course database. Each selection is linked to an appropriate form for entering new data, thus the code for this form is fairly minimal.

“Change Setting” offers the following tables to be updated:

- Instructor
- Departments
- Semesters
- Courses

Currently these options are available for automatic updates as they are the most dynamic tables. There are other tables in the On-Course database that may be edited as needed through the database itself such as semesters, value statements, and evaluation groups.



Add Course

Semester*		Instructor_ID*	
Department Code*		Credit Hours	
Course_ID		*If new Instructor, Department, or Semester, those tables must first be updated	
Section			
Main Menu		Add Course	

Add Instructor
frmAddInstr.frm

Add Departments
frmAddDept.frm

Add Semester
frmAddSemester.frm

Add Course
frmAddCourse.frm

(Add Course (frmAddCourse.frm) is shown above)

When a user selects an option from the “Change Settings” screen, a form is brought up that will allow the user to enter data into a table of the On-Course database.

Some of the selections have drop down combo boxes where information must be chosen from data already in a database such as department code or instructor ID. If a course is being added with a new instructor, semester, or department; the instructor ID, department code, and semester must be entered in their own database prior to creating a course with this information.

Each combo or text box on the form should be filled out to ensure that the database has all the required information to process a course evaluation.

Once the data has been entered, the “Add” button will then save the information into the appropriate table of the On-Course database. The user can then enter another update or click “Main Menu” to return to the Main Form.

Change Setting Forms

Sample Code for frmAddInstr.frm

```
Private Sub cmdAddInstructor_Click()
Dim cnn As New ADODB.Connection
Dim rs As New ADODB.Recordset
Dim cnstr As String
Dim sqlStr As String
Dim returnVal As Integer

'FILL INSTRUCTOR TABLE IN DB
cnstr = "Driver={Microsoft Access Driver (*.mdb)};Dbq=c:\OnCourse\oncourse.mdb;Uid=;Pwd="
With cnn
    .Open cnstr
End With

returnVal = 0
sqlStr = "Select * from Instructor where Instructor_ID='XXXXXXXXXXXX'"

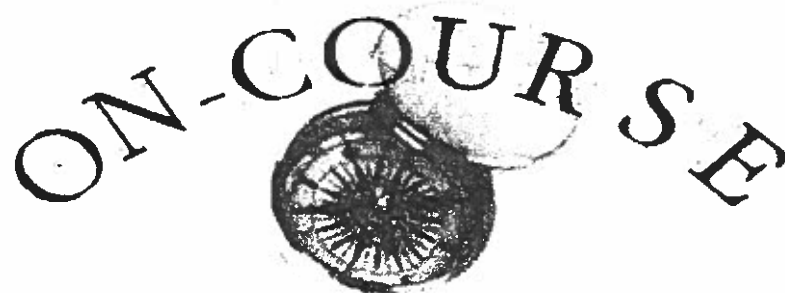
With rs
    .CursorType = adOpenForwardOnly
    .CursorLocation = adUseClient
    .LockType = adLockOptimistic
    .Open sqlStr, cnn
End With

If rs.RecordCount = 0 Then
    'This stores the info into the Instructor database
    rs.AddNew
    rs.Fields("Instructor_ID").Value = Trim(frmAddInstr.txtInstructorID.Text)
    rs.Fields("First_Name").Value = Trim(frmAddInstr.txtFirstName.Text)
    rs.Fields("Last_Name").Value = Trim(frmAddInstr.txtLastName.Text)
    rs.Fields("Department_Code").Value = Trim(frmAddInstr.cmbDept.Text)
    rs.Update
End If

rs.Close
cnn.Close

Unload Me
Load frmAddInstr
frmAddInstr.Show

End Sub
```



Microsoft Access oncourse.mdb

Tables

CourseList	Stores all course information
	Semester, Department_Code, Course_ID, Section, Instructor_ID, Credit_Hrs
Data	Stores all student evaluation data automatically from frmEval.frm
	Course, Student, Code, Box_Number, Row_Number
Data Template	Makes sure there is room for at least 30 student evaluations per class – can be adjusted
	CodeN, BoxN, StudentN, Course
Department	Stores all department information
	Department_Code, Department_Name, Dept_Head_ID
DesiredDataResults	Stores the ideal evaluation results as determined by the department or university
	Code, Box_Number
EvaluationGroup	Stores which departments belong to which evaluation groups. Currently all belong to 1 but can be changed in the future if a variety of value statements are desired
	Evaluation_Group, Department_Code
Instructor	Stores all instructor information
	Instructor_ID, First_Name, Last_Name, Department_Code
Semester	Stores all semester information
	Semester
Statements	Stores all value statements
	Evaluation_Group, Value_ID, Value_Description

ON-COURSE

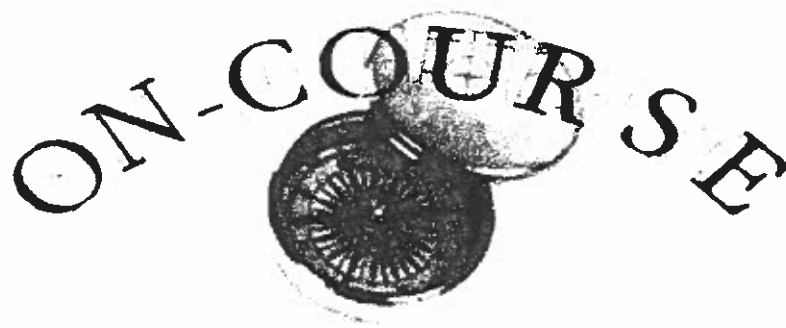
Queries

ActualVsDesiredQuery	Pulls information from DesiredData table, EvalReportQuery#1, and InstructorQuery to give the actual mean for each statement compared to the desired value for each course Course, Code, Mean, DesiredValue, Offset, Value_Description, Instructor_Name
AverageByItem&Course	Lists each value statement and the actual mean for each statement for all classes evaluated Code, Value_Description, & Mean for each class
DataConversion	Converts data from text fields entered from VB to number fields that can be used for calculations StudentN, BoxN, CodeN, Course, Instructor_Name
DataQuery	Union between DataConversionQuery and DataTemplate Table Code, Box Number, Student, Course
EvalReportQuery#1	Lists where each student put each value statement for a particular class and the mean for each statement Course, Instructor_Name, Code, Student (1-n), Mean, Value_Description, Evaluation_Group
InstructorQuery	Combines first & last name for Instructor to make Instructor_Name & Department_Code, Course_ID, and Section to make Course Instructor_Name, Course
MeansByCourseAndItem	Gives average box_number value for each item from a particular course Course, Code, AvgofBox Number
Names	Combines instructor first& last name to make full_Name and links it to Instructor_ID Full_Name, Instructor_ID
StudentCrosstab	Lists course evaluated, and value each student gave for each value statement Course, Code, Student(1-n)



Reports

ActualVsDesiredReport	Lists the value (1-32), the mean for each value, the desired value, the difference between the actual and desired, and the value statements for each course
CourseEvaluationSummary-ByItem	Lists the value (1-32), the students (1-n), where each student placed each value in the evaluation, the mean for each value, and the value statements for each course – Sorted by Value Statements
CourseEvaluationSummary-ByMean	Lists the value (1-32), the students (1-n), where each student placed each value in the evaluation, the mean for each value, and the value statements for each course – Sorted by Mean



Macros

WriteSummarytoExcel	Writes the AverageByItem&Course Query out to Microsoft Excel, saving it as an .xls
WriteTabletoExcel	Writes the ActualVsDesiredQuery out to Microsoft Excel, saving it as an .xls

OnCourse - ActualVsDesired Offset Values

ART 433 01

Aaron Baldwin

Item	Mean	DesiredValue	Offset	Item Description
31	6.33	1	5.33	Teacher is intimidating, condescending, uncompromising
26	6.67	2	4.67	Instructor speaks in monotone voice
19	7.00	3	4.00	Lectures too much
11	5.00	2	3.00	Class is slow, tedious, boring
25	6.33	4	2.33	Enjoyable and funny
30	7.33	5	2.33	Class participation is encouraged
22	5.00	3	2.00	Too much time spent on simple issues
15	4.67	3	1.67	Tests are easy
28	5.67	4	1.67	Workload is heavy
17	5.33	4	1.33	Tests/assignments are returned in a reasonable time
18	4.00	3	1.00	Teaches from the book
1	6.00	5	1.00	Tests related to what is covered in class
32	7.00	6	1.00	Instructor applies subject to the real world
29	7.00	6	1.00	Teacher is well prepared for each class
14	5.00	4	1.00	Tests are mostly essay and short answer
12	5.67	5	0.67	Instructor is punctual and attends class regularly
3	2.67	2	0.67	Talks over your head
13	4.33	4	0.33	Must take good note to be successful
7	5.00	5	0.00	Professor is always available to help
16	6.33	7	-0.67	I learned a lot
5	4.00	5	-1.00	Instructor comments on written work were useful
2	4.00	5	-1.00	Must do homework and be prepared for class
21	3.67	6	-2.33	Professor is fair
27	4.33	7	-2.67	Thought provoking
6	4.00	7	-3.00	Instructor expectations are clear and consistent
24	3.00	6	-3.00	Challenging and demanding
10	3.00	6	-3.00	Clear, well structured presentations
9	4.67	8	-3.33	Teacher sets high academic standards
20	4.67	8	-3.33	Stimulates interest in the subject
4	3.33	7	-3.67	Professor knows subject matter
23	5.33	9	-3.67	Profesor treats students with respect
8	3.67	8	-4.33	Instructor explains concepts well

OnCourse - Course Evaluation Summary - By Item

Course: COIN 325 91

Instructor: James Roberts

-- Student --

Item	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	Mean	Value_Description
1	5	5	5	8	5	8																					6.00	Tests related to what is covered in class
2	1	4	5	4	6	6																					4.33	Must do homework and be prepared for class
3	3	5	4	3	3	4																					3.67	Talks over your head
4	5	5	4	5	8	5																					5.33	Professor knows subject matter
5	4	7	9	6	9	5																					6.67	Instructor comments on written work were useful
6	6	2	1	1	8	2																					3.33	Instructor expectations are clear and consistent
7	4	9	3	4	5	3																					4.67	Professor is always available to help
8	5	3	3	7	7	6																					5.17	Instructor explains concepts well
9	4	6	3	8	2	3																					4.33	Teacher sets high academic standards
10	6	4	5	8	5	4																					5.33	Clear, well structured presentations
11	4	3	7	6	4	1																					4.17	Class is slow, tedious, boring
12	4	5	2	4	1	7																					3.83	Instructor is punctual and attends class regularly
13	2	1	2	5	5	7																					3.67	Must take good note to be successful
14	5	6	6	3	4	5																					4.83	Tests are mostly essay and short answer
15	2	4	7	5	4	8																					5.00	Tests are easy
16	3	5	6	2	6	9																					5.17	I learned a lot
17	6	3	8	3	3	6																					4.83	Tests/assignments are returned in a reasonable time
18	8	3	6	5	4	3																					4.83	Teaches from the book
19	7	2	7	7	2	3																					4.67	Lectures too much
20	7	4	4	6	7	7																					5.83	Stimulates interest in the subject
21	6	4	8	5	7	4																					5.67	Professor is fair
22	9	5	4	2	2	2																					4.00	Too much time spent on simple issues
23	2	2	2	2	6	4																					3.00	Professor treats students with respect
24	8	6	5	3	3	2																					4.50	Challenging and demanding
25	5	8	4	4	8	5																					5.67	Enjoyable and funny
26	3	7	3	7	5	8																					5.50	Instructor speaks in monotone voice
27	5	6	5	4	4	6																					5.00	Thought provoking
28	6	8	5	6	6	7																					6.33	Workload is heavy
29	7	6	7	7	5	5																					6.17	Teacher is well prepared for each class
30	3	7	6	5	3	6																					5.00	Class participation is encouraged
31	7	8	8	6	7	5																					6.83	Teacher is intimidating, condescending, uncompromising
32	8	7	6	9	6	4																					6.67	Instructor applies subject to the real world

OnCourse - Course Evaluation Summary - By Mean

Course: COIN 325 91

Instructor: James Roberts

-- Student --

Item	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Mean	Value_Description
31	7	8	8	6	7	5																			6.83	Teacher is intimidating, condescending, uncompromising
5	4	7	9	6	9	5																			6.67	Instructor comments on written work were useful
32	8	7	6	9	6	4																			6.67	Instructor applies subject to the real world
28	6	8	5	6	6	7																			6.33	Workload is heavy
29	7	6	7	7	5	5																			6.17	Teacher is well prepared for each class
1	5	5	5	8	5	8																			6.00	Tests related to what is covered in class
20	7	4	4	6	7	7																			5.83	Stimulates interest in the subject
21	6	4	8	5	7	4																			5.67	Professor is fair
25	5	8	4	4	8	5																			5.67	Enjoyable and funny
26	3	7	3	7	5	8																			5.50	Instructor speaks in monotone voice
4	5	5	4	5	8	5																			5.33	Professor knows subject matter
10	6	4	5	8	5	4																			5.33	Clear, well structured presentations
8	5	3	3	7	7	6																			5.17	Instructor explains concepts well
16	3	5	6	2	6	9																			5.17	I learned a lot
27	5	6	5	4	4	6																			5.00	Thought provoking
15	2	4	7	5	4	8																			5.00	Tests are easy
30	3	7	6	5	3	6																			5.00	Class participation is encouraged
18	8	3	6	5	4	3																			4.83	Teaches from the book
17	6	3	8	3	3	6																			4.83	Tests/assignments are returned in a reasonable time
14	5	6	6	3	4	5																			4.83	Tests are mostly essay and short answer
7	4	9	3	4	5	3																			4.67	Professor is always available to help
19	7	2	7	7	2	3																			4.67	Lectures too much
24	8	6	5	3	3	2																			4.50	Challenging and demanding
9	4	6	3	8	2	3																			4.33	Teacher sets high academic standards
2	1	4	5	4	6	6																			4.33	Must do homework and be prepared for class
11	4	3	7	6	4	1																			4.17	Class is slow, tedious, boring
22	9	5	4	2	2	2																			4.00	Too much time spent on simple issues
12	4	5	2	4	1	7																			3.83	Instructor is punctual and attends class regularly
13	2	1	2	5	5	7																			3.67	Must take good note to be successful
3	3	5	4	3	3	4																			3.67	Talks over your head
6	6	2	1	1	8	2																			3.33	Instructor expectations are clear and consistent
23	2	2	2	2	6	4																			3.00	Professor treats students with respect

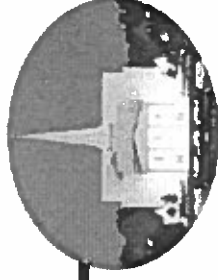
Senior Project Defense

May 9, 2001



About CSU

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Jared Dingus
Sandeep Reddy
Paul Stubblefield

Senior Project Defense

May 9, 2001

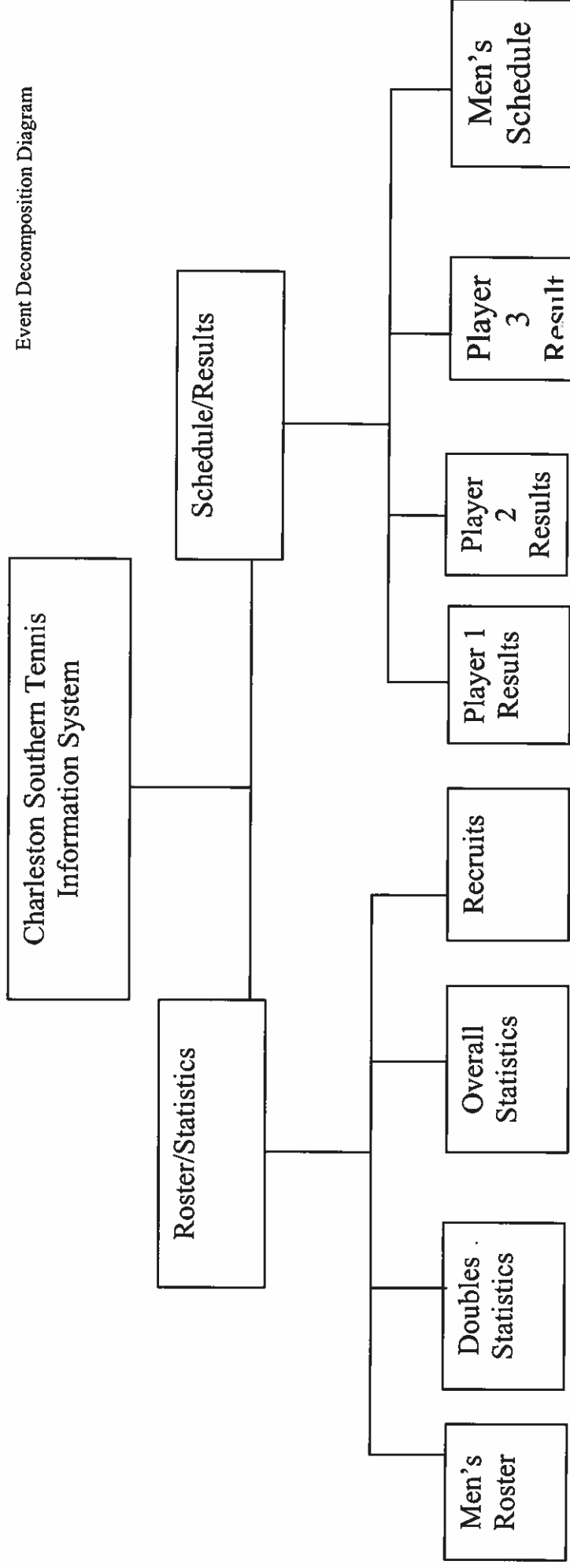
Jared Dingus
Sandeep Reddy
Paul Stubblefield

On January 15, 2001, Coach Hans Olsen from the Charleston Southern University Tennis Team contacted the CSU student consultants in regards to a project that would greatly enhance the coaching staff use. CSU student consultants will create a database program for the CSU men and women's tennis teams. This database will contain the names of the tennis players, their results, player statistics, and recruiting information. This database program would be used and can be updated the coaching staff for the teams purpose.

The potential benefits of this service will be immediate. The staff could use all the information of the CSU Tennis program. This information includes the personal information of the tennis players in the team, their individual results, teams schedule and results, and recruit information.

The project feasibility assessment is as follows. The database will be created for both the men and women's teams. The database will consist of statistics, bios, schedules, results, etc. CSU Student Consultants recommend that this database is to be implemented on the internet. The overall goal of this project is to have a private database management system for the coaches in order for easy updates. The next step is to try to implement the system on the Internet in order for the coaches to make changes after each match despite their location.

Event Decomposition Diagram



COMPLETE TABLE DEFINITONS

MENS_TENNIS_TEAM_ROSTER

PK2 CSU_ID_Number; 9 Characters, Numerical, mask #####
FK Player_name; Alphabetical, 20 Characters
Classification; Alphabetical, 10 Characters
Country; Alphabetical, 20 Characters

MENS_OVERALL_STATS

PK2 CSU_ID_Number; 9 Characters, Numerical, mask #####
FK Player_name; Alphabetical, 20 Characters
Number_1; 10 Characters, Memo
Number_2; 10 Characters, Memo
Number_3; 10 Characters, Memo
Number_4; 10 Characters, Memo
Number_5; 10 Characters, Memo
Number_6; 10 Characters, Memo
Winning_%; 5 characters, Memo

MENS_DOUBLES_STATS

PK2 CSU_ID_Number; 9 Characters, Numerical, mask #####
FK Player_name; Alphabetical, 20 Characters
Number_1; 10 Characters, Memo
Number_2; 10 Characters, Memo
Number_3; 10 Characters, Memo

RECRUITS

PK1 Player_name; Alphabetical, 20 Characters
Country; Alphabetical, 20 Characters
High_school; Alphabetical, 20 Characters
Classification; Alphabetical, 10 Characters
Telephone_Num; 15 Characters, Numerical

MENS_SCHEDULE

PK1 Date; 10 Characters, Date
Opponent_School; 30 Characters, Alphabetical
Match_result; 5 Characters, Alphabetical

SANDEEP_REDDY

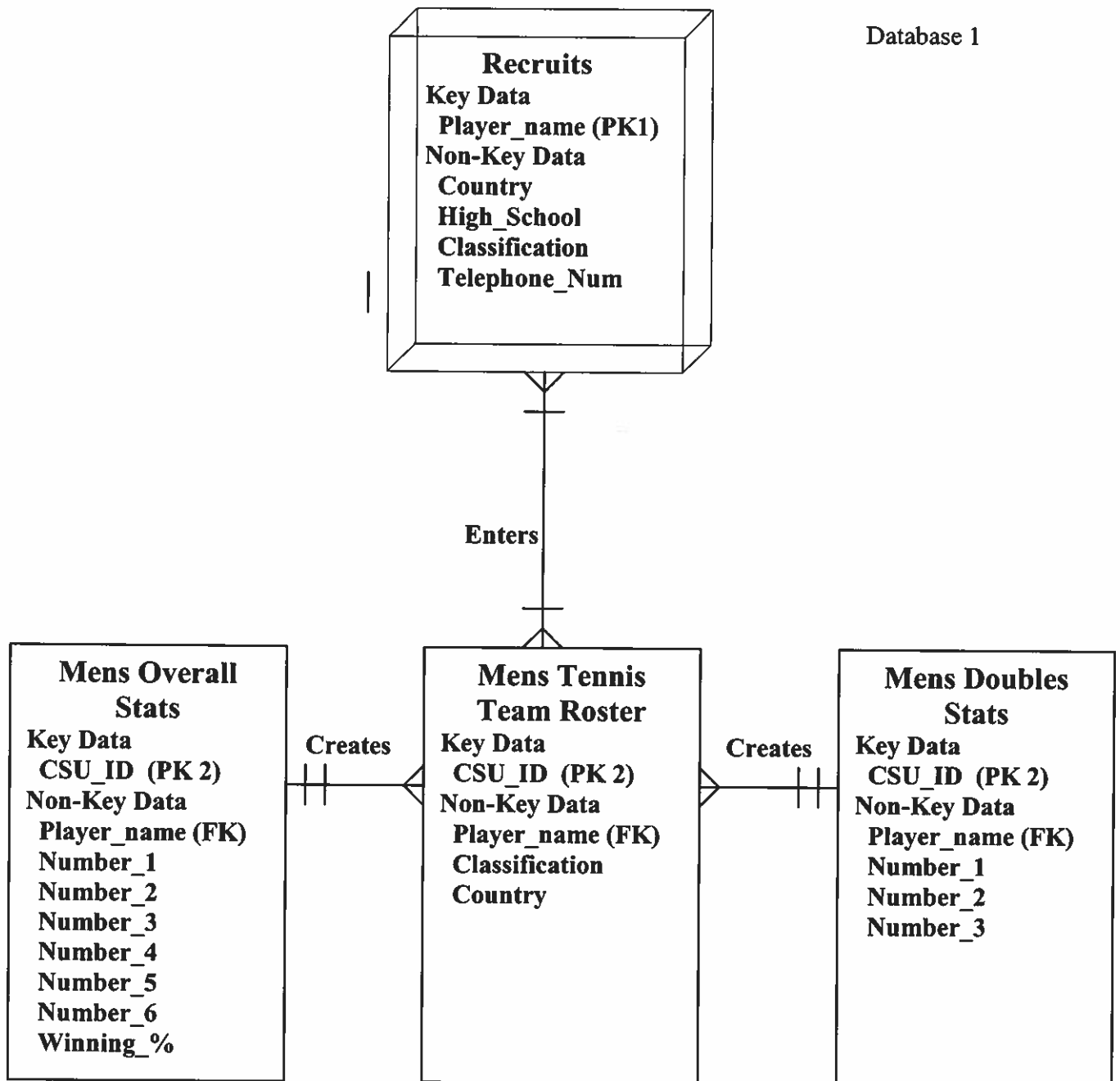
PK2 Match_No; 3 Characters, Numerical
FK Date; 10 Characters, Date
Opponent; 20 Characters, Alphabetical
Opponent_School; 30 Characters, Alphabetical
Match_result; 2 Characters, Alphabetical
Score; 5 Characters, Memo

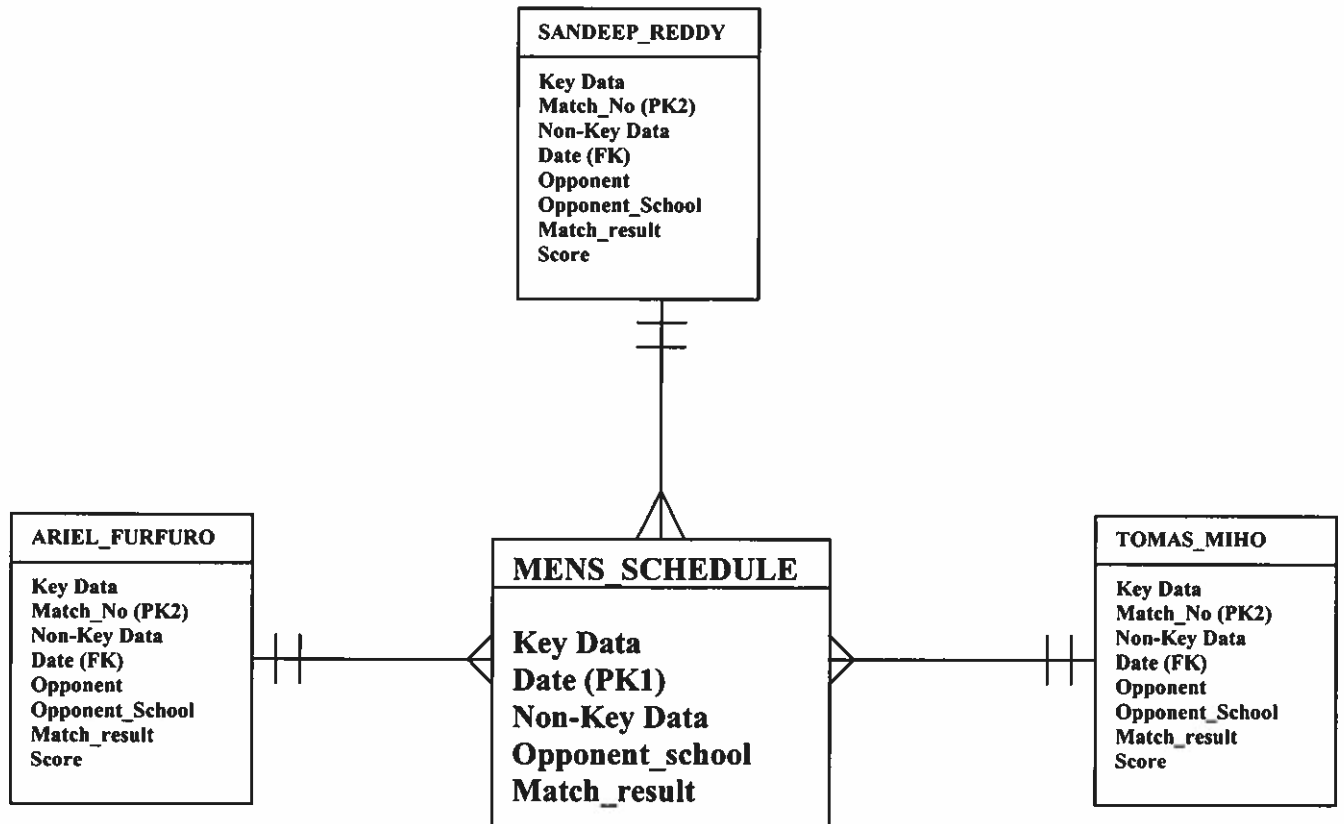
TOMAS_MIHO

PK2 Match_No; 3 Characters, Numerical
FK Date; 10 Characters, Date
Opponent; 20 Characters, Alphabetical
Opponent_School; 30 Characters, Alphabetical
Match_result; 2 Characters, Alphabetical
Score; 5 Characters, Memo

ARIEL_FURFURO

PK2 Match_No; 3 Characters, Numerical
FK Date; 10 Characters, Date
Opponent; 20 Characters, Alphabetical
Opponent_School; 30 Characters, Alphabetical
Match_result; 2 Characters, Alphabetical
Score; 5 Characters, Memo





CSU_ID_Num	Player_Name	Classification	COUNTRY
12847	Falla	Sophomore	COLUMBIA
23423	Furfuro	Junior	ARGENTINA
30993	Miho	Junior	SLOVAK REP
32321	Reddy	Senior	INDIA
39753	Rechag	Freshman	MOROCCO
87354	Withrow	Junior	AMERICA
94755	Wadehra	Junior	INDIA
95483	Villaroel	Sophomore	BOLIVIA

CSU ID	Num	Player Name	Number_One	Number_Two	Number_Three	Number_Four	Number_Five	Number_Six
12847	Santiago Falla	N/A	N/A	N/A	N/A	0-1	1-1	10-4
23423	Ariel Furfuro	N/A	N/A	N/A	N/A	1-1	12-4	N/A
30993	Tomas Miho	11-8	N/A	N/A	N/A	N/A	N/A	N/A
32321	Sandeep Reddy	N/A	17-4	N/A	N/A	N/A	N/A	N/A
39753	Soufiane Rechag	N/A	N/A	N/A	N/A	N/A	N/A	6-0
94755	Rohan Wadehra	N/A	N/A	2-1	N/A	12-4	N/A	N/A
95483	Rodrigo Villaruel	N/A	N/A	11-6	N/A	N/A	N/A	N/A

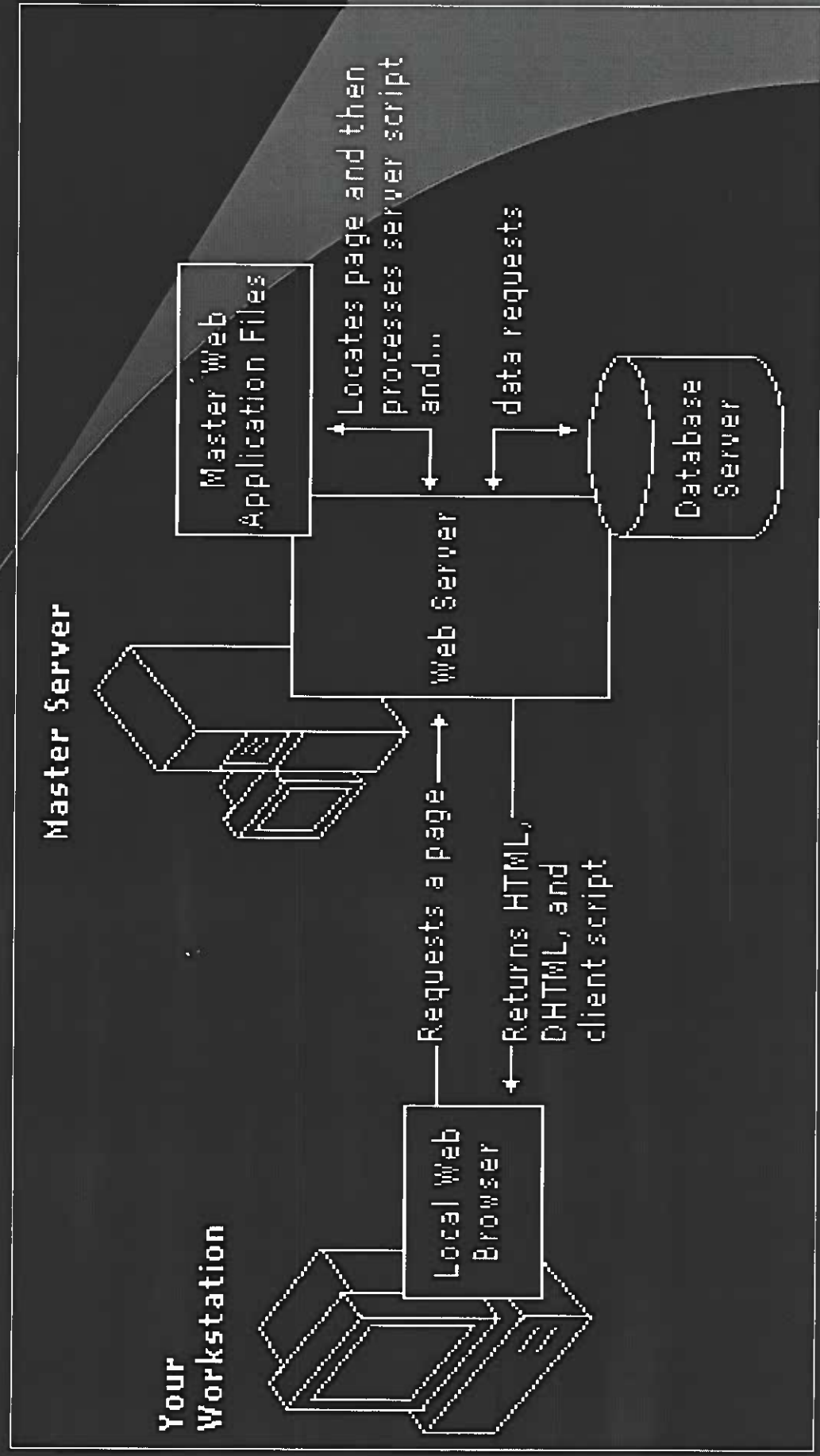
Win_Perc
64.70%
72.22%
57.89%
80.95%
100%
73.68%
64.70%

CSU_ID_Num	Player_Name	Number_One	Number_Two	Number_Three	Win %
12847	Falla/Reddy	0	0	1-0	100%
23423	Furfuro/Falla	0	0	0-1	0%
30993	Miho/Wadehra	14-6	0	0	70%
32321	Reddy/Rechag	0	1-0	12-4	76.47%
39753	Rechag/Falla	0	0	1-0	100%
95483	Villaroel/Furfuro	0	11-7	0	61.1%

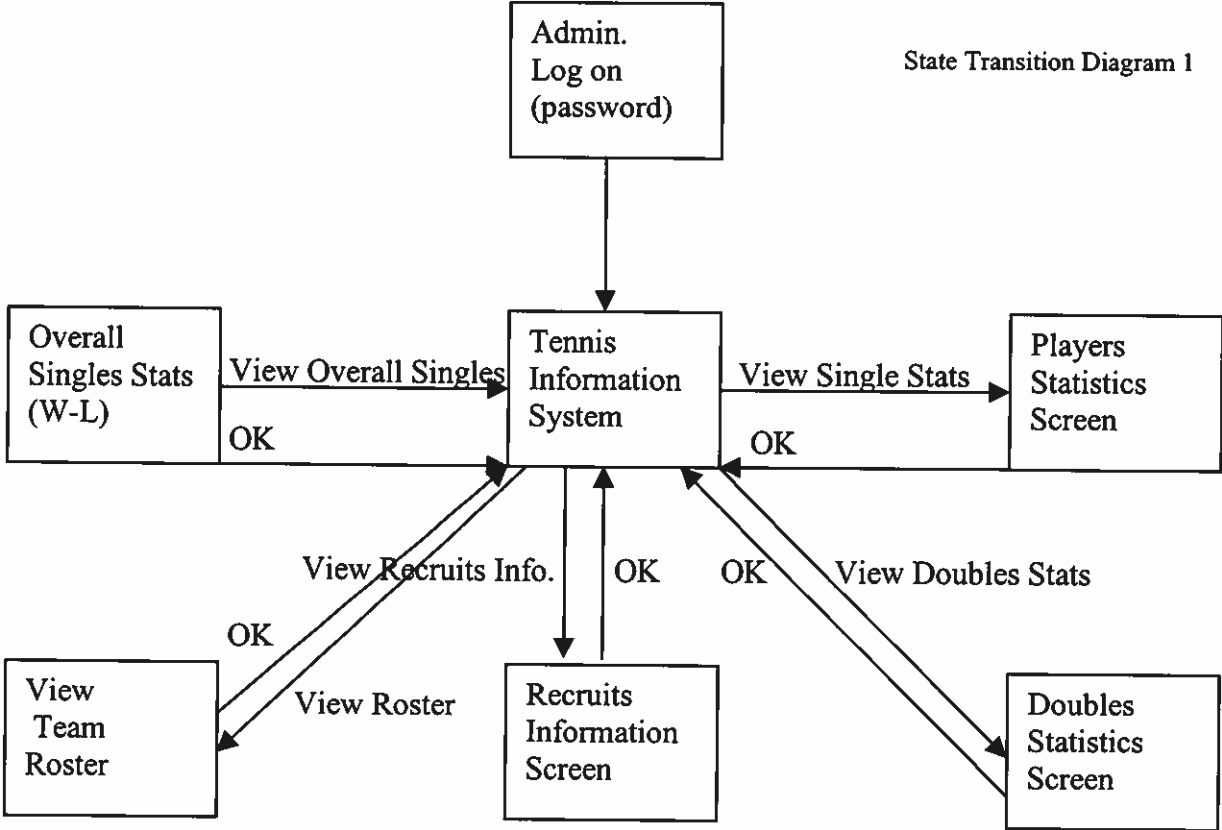
Opponent	Opponent_School	Date_of_Match	SCORE	RESULT
Glynn Hulley	Francis Marrion	2/9/01	6-2, 7-5	W
David Nelson	University of South Carolina	2/10/01	1-6, 2-6	L
Sagi Zakin	College of Charleston	2/14/01	7-6, 6-0	W
Woichek Novak	Georgia Southern	2/17/01	6-4, 6-4	W
Richard Dutton	Elon College	3/2/01	7-5, 6-0	W
Luis Rosado	James Madison	3/7/01	6-3, 6-2	W
Jacob Nordwall	UNC-Charlotte	3/8/01	6-2, 6-2	W
Johan Dorth	High Point Univ	3/16/01	6-0, 6-4	W
Thomas Smid	University of Miami	3/19/01	7-6, 6-4	L
Yalin Bilgin	Florida Atlantic Univ	3/20/01	3-6, 6-2, 5-7	L
Domencio Marra	Univ of Arkansas	3/28/01	4-6, 6-3, 7-6	W
Nick Malone	Brown University	3/30/01	6-1, 6-3	W
Pablo Shagun	East Tennesse State Univ	3/31/01	6-0, 6-2	W
Stephen Paulos	UNC-Ashville	4/4/01	6-1, 6-0	W
Bruno Coelho	Liberty	4/6/01	6-0, 6-4	W
Barthlo Osthuzien	Radford	4/7/01	4-6, 3-6	L
Jean Phillip Cloutier	Coastal Carolina Univ	4/11/01	6-3, 6-0	W
Micheal Kwiecien	Winthrop Univ	4/13/01	6-3, 7-6	W
Stephen Paulos	UNC-Ashville	4/19/01	6-1, 6-0	W
Barthlo Osthuzien	Radford	4/20/01	6-3, 6-0	W
Micheal Kwiecien	Winthrop Univ	4/21/01	6-2, 6-1	W

Date	Opponent	Match_Res
2/9/01	Francis Marion	W 4-2
2/10/01	University of South Carolina	L 0-4
2/14/01	College of Charleston	W 6-1
2/17/01	GEORGIA SOUTHERN	W 6-1
3/2/01	ELON COLLEGE*	W 6-1
3/3/01	HAMPTON	RAIN
3/7/01	JAMES MADISON	W 7-0
3/8/01	UNC-CHARLOTTE	W 4-3
3/12/01	EAST CAROLINA	RAIN
3/16/01	HIGH POINT UNIVERSITY*	W 7-0
3/19/01	Miami University	L 0-7
3/20/01	Florida Atlantic Univ	L 0-7
3/28/01	UNIV OF ARKANSAS (LR)	W 5-2
3/30/01	BROWN UNIVERSITY	W 5-2
3/31/01	EAST TENNESSE STATE	W 4-3
4/4/01	UNC-Ashville*	W 7-0
4/6/01	Liberty*	W 7-0
4/7/01	Radford*	W 4-3
4/11/01	Coastal carolina*	W 6-1
4/13/01	WINTHROP*	W 6-1
4/18/01	BIG SOUH CONFERENCE CHAMPINSHIP#	
4/19/01	UNC ASHVILLE#	W 6-0
4/20/01	RADFORD#	W 4-0
4/21/01	WINTHROP#	W 4-1

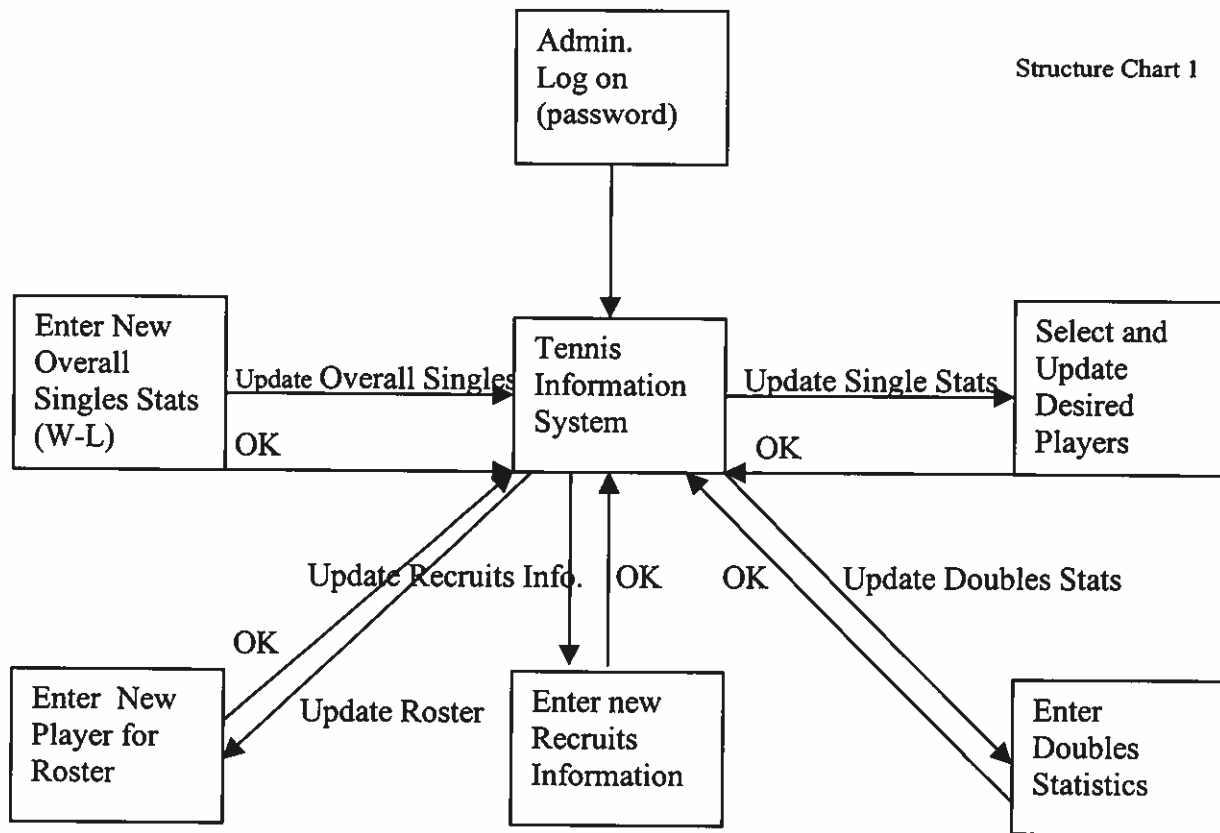
How InterDev Works



State Transition Diagram 1



Structure Chart 1



TENNIS (M)

[Home Page](#)

[Administrator Login](#)

[Update Mens Roster](#)

[View Mens Roster](#)

[Update Mens Singles Stats](#)

[View Mens Singles Stats](#)

[Update Mens Doubles Stats](#)

[View Mens Doubles Stats](#)

[Update Recruits](#)

[View Recruits](#)

[View Overall Singles Stats](#)

[View Overall Doubles Stats](#)

[2001 Men's Schedule](#)



Mens Roster Update

CSU ID Number

Player Name

Classification

Country

32321

Sandeep Reddy

Senior

India



```
//Jared Dingus
//Paul Stubblefield
//Sandeep Reddy
```

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//Update_Roster.asp  Asp Page to Update the Mens_Tennis_Team
_Roster Table
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Benefits/Concluding Remarks

- The coaching staff has easier access to the database, and can update the information more efficiently.
- The coaching staff can update the database from any off campus location via the World Wide Web.
- The coach can personally update the database without the Sports Information Director or without accessing the database itself.